**Name:** **Admission No.**

**Indian Institute of Technology (Indian School of Mines) Dhanbad**

**Department of Mining Engineering**

Endsem Winter Semester 2023-2024 for 4th Year B.Tech. + 1st Year Ph.D. (36 nos.)

**Subject:** Deep Coal Mining (Code: MND 412)

**Time:** 180 minutes (04:00 PM – 07:00 PM) **Date:** 05-05-2024 **Maximum Marks:** 96

*Use of scientific calculator is allowed*

*Answer questions of **all** the three sections and **marks** are assigned against each question*

***Section A: Short Descriptive Answer Type Question (Answer in 1-2 lines/pointwise) – 28 marks***

*Answer all the questions and each question carries 1 mark*

**1.** Draw a general layout of mine subsidence stations above a Longwall/Bord and Pillar panel.

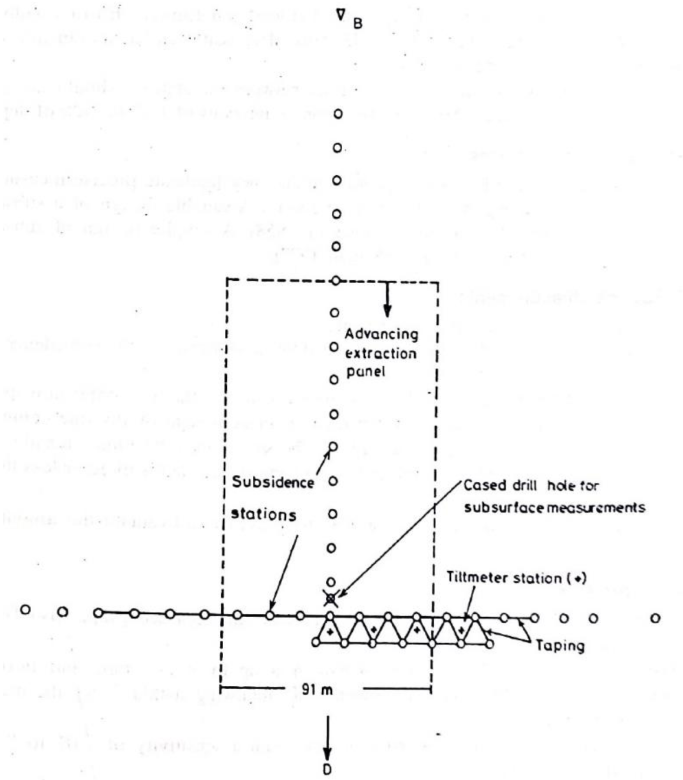



**Figure 13.33 :** Surface instrumentation for subsidence measurement (After Panek, 1972)

**2.** What is the best time to start repairing a building lying in the zone of subsidence trough?

**If mining reaches the reserve limit and comes to a permanent halt, then repairs could begin almost immediately. Otherwise, some assurance that further subsidence will not occur must be obtained before repairs should begin. In this regard, the limit to subsidence ahead of a face or beyond a panel edge is defined by the angle of draw.**

**3.** Mention the average angle of draw for Indian geo-mining conditions.

**Angle of draw Indian coalfields: 4° - 21°.**

**4.** State the limitations of estimating subsidence using SHB of NCB, UK.

**It has been developed considering the geo-mining conditions of coal mines of UK and does not consider the time effects: The period during which mine subsidence occurs**

**5.** Mention the factors upon which peak stresses depends at hole boundaries.

**Peak stresses occur at hole boundaries and depend on:**

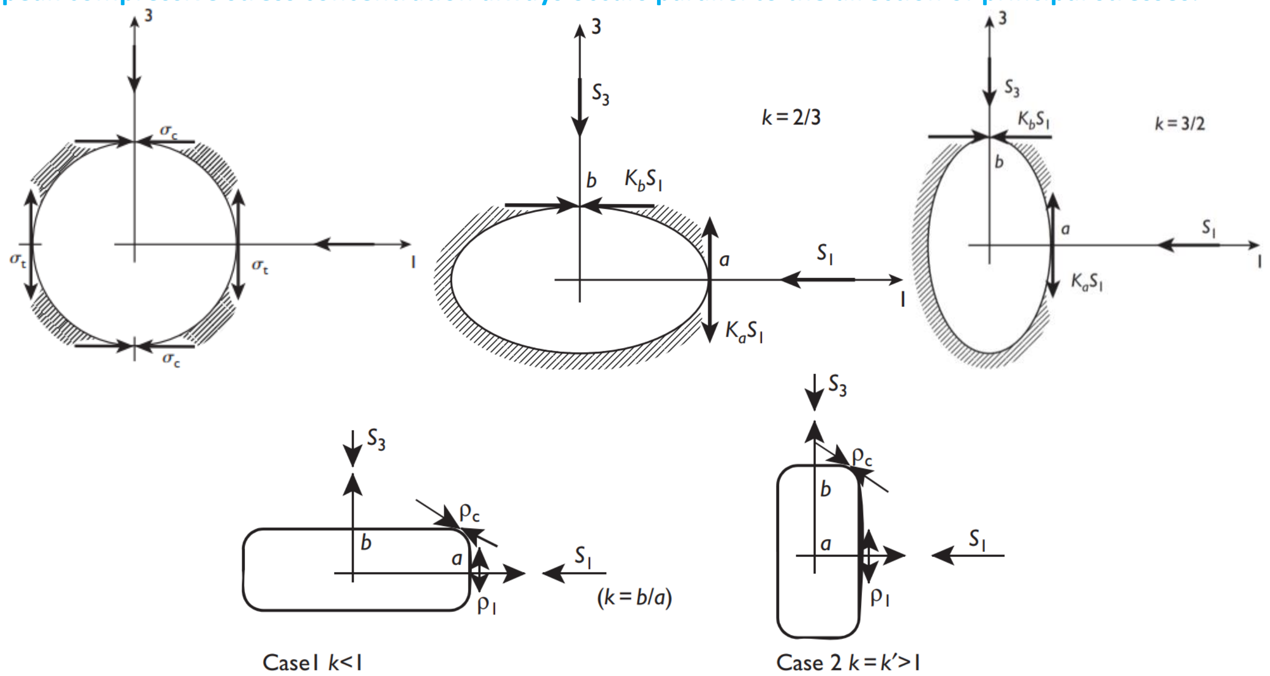
**hole shape**

**aspect ratio**

**principal stress ratio (M)**

**orientation**

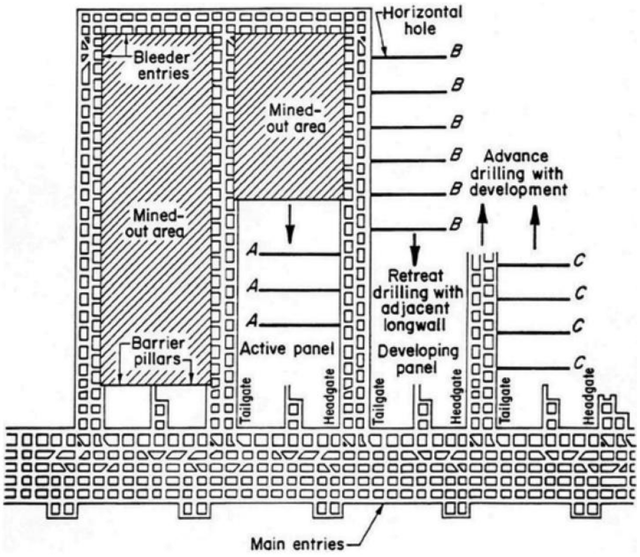
6. Which part of a circular, rectangular and elliptical shaft experiences maximum stress concentration?
**peak compressive stress concentration always occurs parallel to the direction of principal stresses.**



Sketch for rectangular opening stress concentration

7. Discuss the advantages of two-entry over one-entry in terms of inter-panel barrier pillar in longwall panel development.
**Two-entry inter-panel barrier pillar- nitrogen flushing, entries for adjacent panel, goaf can be checked can controlled.**

8. What are the different methane sources at the longwall face and how it is drained out?
**The sources of methane emission at the longwall face are as follows (Trackemas and Peng, 2013):**
**(1) gas released from the coal broken by the shearer,**
**(2) gas emitted from the broken coal on the face conveyor (AFC),**
**(3) gas emitted from the coal transported on the belt conveyor, and**
**(4) background gas emitted from the coal face and from the adjoining ribs in the intake gateroads.**
**To be fully effective in terms of timing and space, methane drainage methods can be divided into two types: pre-mining and post-mining methods. The pre-mining methods consist of (Thakur, 2006):**
**(1) horizontal inseam boreholes,**
**(2) surface stimulated vertical boreholes (frac wells),**
**(3) surface directionally drilled boreholes, and**
**(4) dual well drilling system (Zupanick, 2005).**
**The post-mining method is mainly the vertical gob well method. In practice, a combination of these methods is used to degasify coal seams as much as possible before and during mining to reduce the amount of gas emissions from the gob into the ventilation system. The optimum methane drainage system for a mine depends on the amount of methane produced, emission pattern, and geology of coal seam and surrounding strata. It may need to be evaluated and adjusted on a continuing basis to ensure optimum capture of methane over time.**



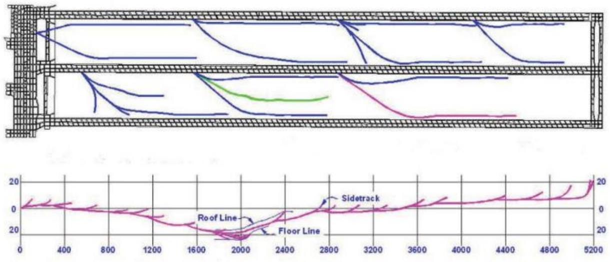




*Figure 11.3.2* Ultra-long holes drilled from gateroads in advance of gateroad developmen
Source: Target Drilling (2006)

**9.** What were the main reasons of instability in gateroads and face of Adriyala Longwall Project of Singareni Collieries Company Limited, Telangana?

**Multiple clay bands and insufficient support system of longer lengths**

**10.** What are the major reasons for not achieving adequate success of longwall mining in India?

**Churcha longwall face failed due to dynamic loading.**
**Jhanjra with shallow depth longwall working face ran into acute spares problem.**
**Khottadih face failed after successful completion of two Longwall panels due to dynamic loading and underrated capacity of supports.**
**GDK11A failed due to underrated capacity of supports.**
**Poor understanding of Indian rock mass behaviour.**
**Indian coal seams are not exactly flat and found with a number of geological discontinuities.**
**Longwall technology is imported and lacks spare parts availability in India.**
**Reluctance of miners and administration.**
**Huge capital cost and high gestation period.**
**Lack of skilled manpower.**
**More damage to surface features due to subsidence**

**11.** Define thick coal seam as per India and other countries.

**Thick coal seams refer to those with a seam thickness greater than 3.5 m in China.**
**In India a thick seam is considered to have thickness more than or equal to 4.5-4.8 m.**

**12.** Discuss the different mining methods for thick coal seams extraction.

**At present, the common mining methods for thick coal seams are**

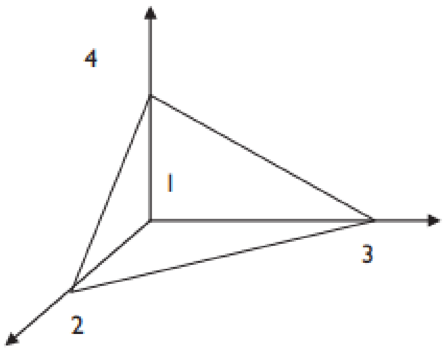
- **Large-cutting-height mining method**
- **Top-coal caving mining: Single-pass full-seam longwall top-coal caving mining**
- **top-coal caving mining underslicing a longwall mining method,**
- **slicing top-coal caving mining method, and**
- **large-cutting-height top-coal caving mining method.**

**13.** What is the minimum number of geophones required to locate an event occurred in 3D space?

**Four**

**14.** An event occurs in 3D-space and is simultaneously recorded by receivers positioned at the vertices of a tetrahedron. Determine the event location and time if it is locatable using minimum number of geophones.

**By the guiding principle**




$$\sqrt{(0-x_s)^2+(0-y_s)^2+(0-z_s)^2}=v(t_o-t_s)$$
$$\sqrt{(x_2-x_s)^2+(0-y_s)^2+(0-z_s)^2}=v(t_o-t_s)$$
$$\sqrt{(0-x_s)^2+(y_3-y_s)^2+(0-z_s)^2}=v(t_o-t_s)$$
$$\sqrt{(0-x_s)^2+(0-y_s)^2+(z_4-z_s)^2}=v(t_o-t_s)$$

**where the receivers sense the event at the same time $t_o$.**

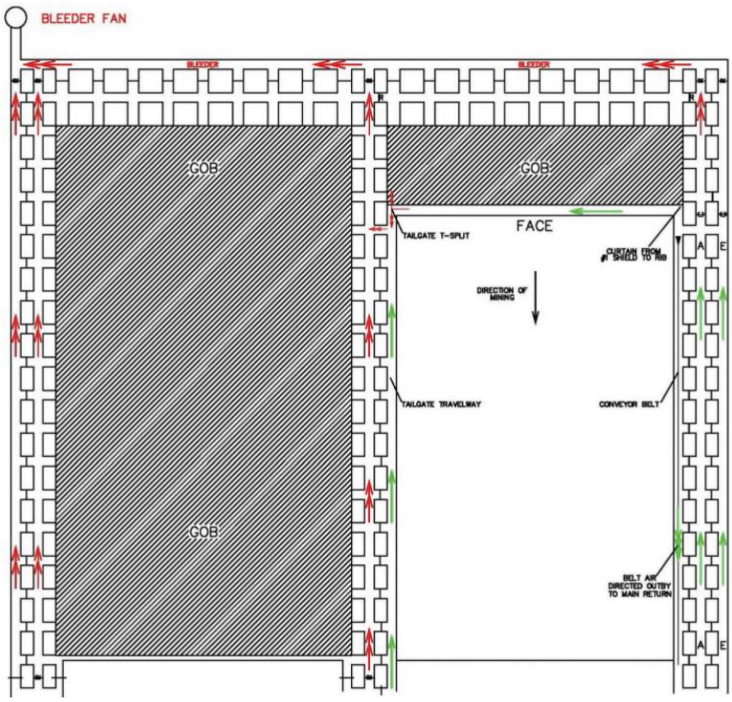
**The solution to this system for the source location using the first and second, then the first and third, and finally the first and fourth equations is $x_s=\frac{x_2}{2}, y_s=\frac{y_3}{2}, z_s=\frac{z_4}{2}, .$**

**Substitution of this result back into the first equation gives $t_s=t_o-\frac{1}{4}v\sqrt{(x_2)^2+(y_3)^2+(z_4)^2}$**

**15.** What are the ventilation challenges for deep coal mines?

**methane and dust concentration**
**high temperature**
**leakages**
**spontaneous heating**
**nitrogen flushing**

**16.** Discuss bleeder ventilation system with their advantages and disadvantages.

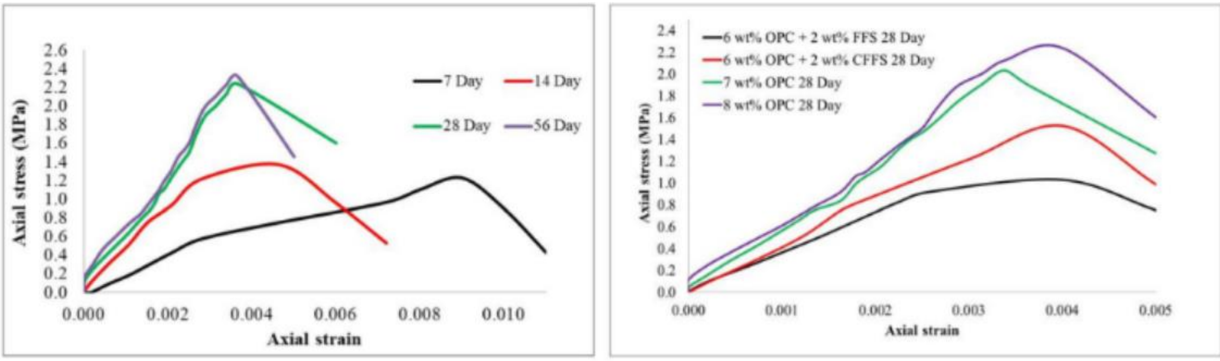



**Bleeder entries are connected to gob areas at strategic locations to do the following:**
**(1) control airflow through gobs,**
**(2) induce gob gas drainage, and**
**(3) minimize the hazard of gob gas expansion into fresh air.**

**One advantage of using a bleeder system that takes fresh air from the main entries and directs it toward the bleeder fan is that it is easier to control the air and keep it moving primarily in one direction only except that the air on the conveyor belt is usually directed outby. Otherwise, all air moves from outby to inby. Another longwall ventilation system makes use of the main return located in the main entries outby the longwall face. Most of the longwall air flows outby, and only a small amount of air makes it to the bleeder entries compared to the previous system that has all air reporting to the bleeder. However, since longwall mining is a type of retreat mining, MSHA requires a bleeder system behind the gob inby the longwall face. The bleeder entries may be tied to the main return or may have a bleeder fan. In either case, the quantity of air flowing through the bleeder entries is much less than the previous bleeder system, as most of the air from the longwall face is directed outby in the tailgate entries.**

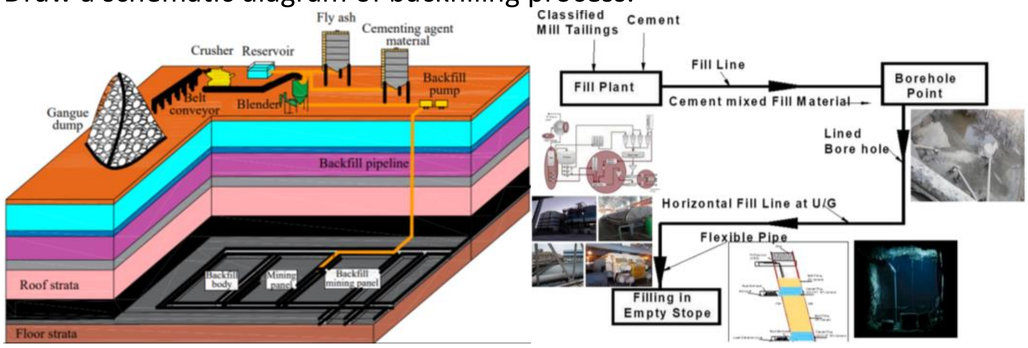
**17.** Define slump test and its significance in paste backfill.
to evaluate the "workability or consistency" of fresh concrete. The slump test arises from the requirement of determining the flowability of fresh concrete. If the concrete is too thick, the mixture may not flow into the tight corners of the structures to be constructed. Conversely, if the concrete has higher water content, the flowability is improved, but the strength of the final hardened concrete is reduced. Therefore, it is required to maintain a balance between the flow properties. Similarly, during paste backfilling, transportation of the paste is one of the critical elements. It is related to the workability of paste backfill. An optimum solid proportion of paste mix facilitates the bulk disposal of waste material with optimum energy consumption. Paste backfill with high water content is easily flowable but may not give the required strength. Therefore, optimum water percentage of paste backfill is maintained. The slump test provides a quick and onsite tool for maintaining the optimum solid proportion where sophisticated instruments are not available. A quick and easy measurement method to determine the consistency of the freshly prepared paste is the slump test. Paste backfill behaves as a Bingham fluid, which means that it needs an applied force to commence the flow. The flowability of paste fill is dependent on the yield stress and viscosity of the paste material. The yield stress of the material is related to the slump of paste. The slump is an experimental value, which is dependent on the type of material, paste density, slumping speed and yield stress of paste. Previous studies showed that the slump test could be used for predicting the yield stress of paste. Yin et al. studied the effect of solid proportions on the rheology of cemented paste backfill (CPB) prepared using lead–zinc tailings. The authors' determined 78 wt% as the critical value of solid content in paste backfill. More recently, Li et al. studied the correlation between the spread and yield stress of paste fill using tailings from Chinese and Indonesian mines and determined the yield stress of the paste in the range of 115–385 kPa. Clayton et al. have proposed that paste with a slump of 190–200 mm would be workable. Similarly, Behera et al. suggested a slump of 195 mm for paste backfill prepared using lead–zinc mill tailings and determined the optimum solid percentage of paste as 77%.

**18.** Discuss the effects of curing time and binder type and its percentage on stress-strain behavior of paste backfill.




**19.** Define paste as a suitable backfill material.
Paste is a mixture of material, which akin to toothpaste, comprising 75–85% solid by weight and 1 wt% to 12 wt% binder, containing at least 15 wt% solid fraction <20 μm, when poured into voids the backfill does not bleed water, does not settle in the pipeline, and has a slump<250 mm

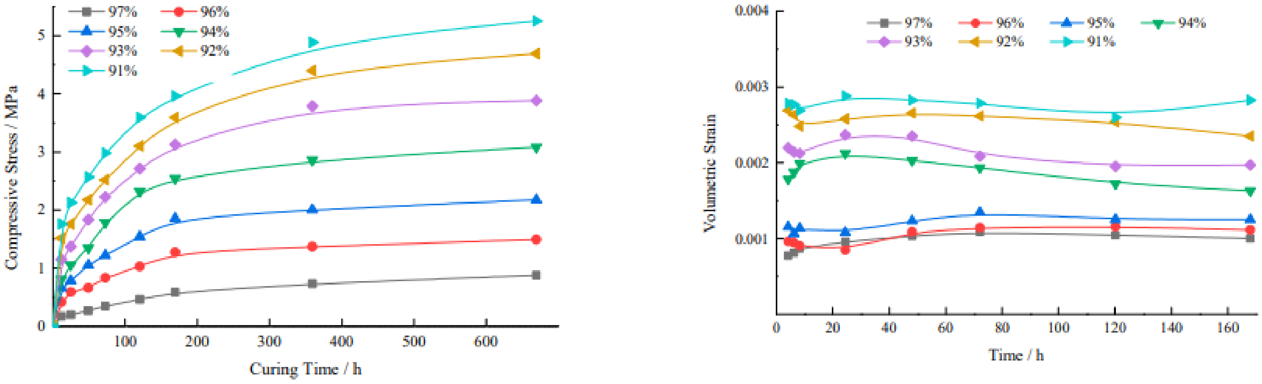
**20.** Draw a schematic diagram of backfilling process.




**21.** Discuss the significance of initial and final settling time of paste backfill.
Setting time is the time taken by the weakly bound paste backfill ingredients for hardening.

Vicat needle apparatus (penetration resistance method) is typically used to determine the initial setting time (IST) and final setting time (FST) of paste at room temperature, according to IS: 4031 (Part 5) (1988).
IST is the "time the needle fails to penetrate the test specimen by 5.0 ± 0.5 mm, measured from the bottom of the paste mould".
FST is determined as the time at which the needle could not noticeably penetrate the paste (i.e., the specimen has almost become a solid structure).
Rai et al. and Yang et al. stated an increase in setting time of paste backfill using slag as a binder replacement. Further, Singh et al. stated an initial setting time (IST) of 9–13 h and a final setting time (FST) of 20–24 h of paste backfill using lead–zinc mill tailings, fly ash and jarosite.
The backfill structure should have sufficient strength to sustain the increased load due to stress redistribution and it should not fail during the extraction of adjacent secondary stopes.
With the progress of time, the loose backfill gains strength and its geomechanical properties changes, which can be investigated by the uniaxial compressive strength (UCS), Young's modulus/stiffness, tensile strength and cohesion.

**22.** Discuss the stress-strain behavior with respect to water content in backfill materials.




(a) (b)

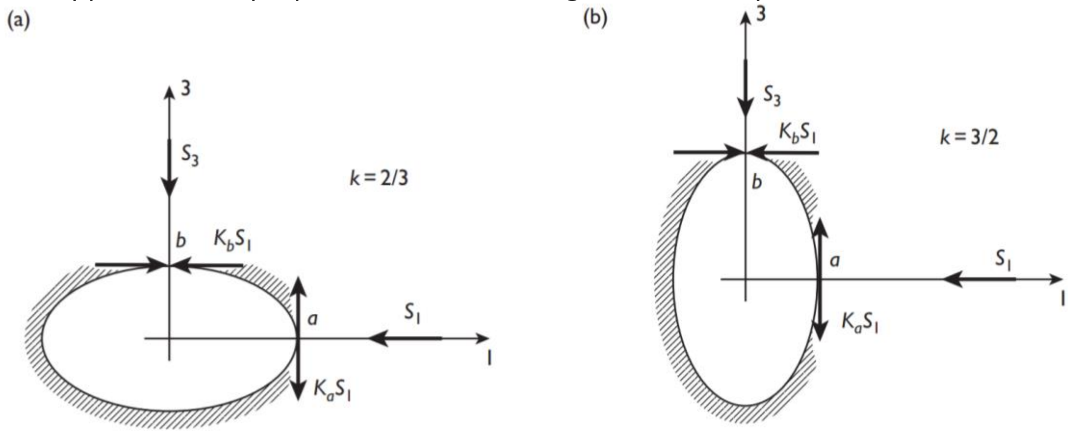
**23.** Differentiate between main weighting and periodic weighting in Longwall Mining Methods.
After face advancing, the roof layers start to sag and separate from their upper layers. In this case, the layer acts like a simple beam, in which the head of beam is located on the face and powered support, and the other side is on the barrier pillars. The load is applied to the beam. After face advancing, the length of the beam and the pressure on the two heads of beam increase. Overhanging of this beam depends on the mechanical properties and rock type. This beam falls at a certain distance from the face in the middle of its span due to the development of shear joints and weight of upper layers, which is known as the first or main roof weighting.
After the first failure, the beam acts as a cantilever beam and remains an overhang on the back of the support system and falls periodically after face and support system advancing. This stage of roof falling is called the periodic weighting.

**24.** Why 2-legged powered roof supports are more popular than 4-legged?
A chock shield (4-legged) has the highest supporting efficiency and is suitable for a hard roof.
Usually, front legs take more load than rear legs due to premature caving nature of immediate roof. Unused rear two legs were abandoned gradually.

**25.** Classify the different nature of roof and their weighting based on Caveability Index.

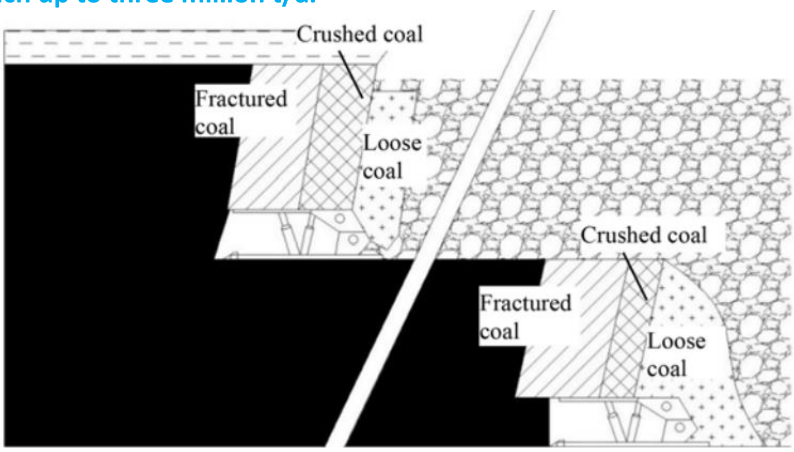
| Category | Nature of Caving | Range of Cavability Index | Main fall span, $S_m$ (m) | Weighting Dynamism |
|---|---|---|---|---|
| Category- I | Easily cavable | I ≤2000 | $S_m < 35$ | - |
| Category- II | Moderately cavable | 2000 < I ≤ 5000 | $35 \leq S_m < 55$ | - |
| Category- III | Cavable with difficulty | 5000 < I ≤ 10,000 | $55 \leq S_m < 80$ | May or may not be |
| Category- IV | Cavable with substantial difficulty | 10000 < I ≤ 14000 | $80 \leq S_m < 95$ | May or may not be |
| Category- V | Cavable with extreme difficulty | I ≥ 14000 | $S_m \geq 95$ | Dynamic |

**26.** Draw locations of peak tensile and compressive stress concentration about an elliptical hole **(a)** when the applied load is parallel to the long axis of an ellipse with semi-axes ratio b/a = 2/3 and **(b)** when the applied load is perpendicular to the long axis of an ellipse with semi-axes ratio b/a = 3/2.




**27.** Discuss about a suitable underground coal mining method to extract a 30 m thick coal seam.

**Slicing top-coal caving mining method: When the coal seam is more than 20 m thick, multiple tens of meters or even up to hundreds of meters, several slicing top-coal caving mining panels can be employed. The coal seam is divided into several slices 10 to 15 m thick from roof to floor and mined in sequential order by the top-coal caving mining technology (Fig.). It also can be used in steeply inclined coal seams more than 20 m thick (Wang, 2009). In order to improve the recovery ratio of top coal, a wire mesh is laid on the floor when the first slice mining is being conducted, putting the mining operation of all lower panels be done under the wire mesh, reducing the wastes and increasing the recovery ratio. Shenhua Xinjiang Energy Co. Ltd. Wudong mine adopts the slicing top-coal caving mining in steeply inclined thick coal seams (Li, 2015). Its average dip angle is 87 degrees and the average thickness is 37.45 m. At present, surface mining has reached the 1500 m level (1850 m on) (Fig. 12.11). The panel length is the horizontal thickness of the coal seam. Panel production can reach up to three million t/a.**




**28.** Discuss large-cutting-height mining method and issues associated with this method.

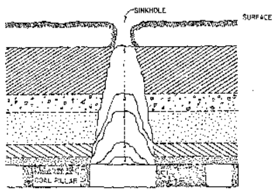
**The large-cutting-height mining is referred to those longwall panels that extract coal more than 3.5 m thick and the support height of the shields is usually more than 3.8 m. At present, the maximum support height of the large mining height panels is 8.8 m, generally 5 to 7 m, with the maximum annual production 15 Mt, generally 6 to 10 Mt. The special features of the fully mechanized large-cutting-height longwall mining are that the shield supports are very high, the shearer has large power capacity, the carrying capacity of AFC (armored chain conveyor) is also very large, the roadways are wide, and auxiliary equipment is also heavy-duty. This mining method is used for the thick coal seams, the uniaxial compressive strength (UCS) of which is greater than 20 MPa, and fractures are not well-developed. The primary issues for the large-cutting-height mining method are to control face spalling and roof weighting at the face.**

*Section B: Short Descriptive Answer Type Question (Answer in 3-4 lines/pointwise) – 20 marks*

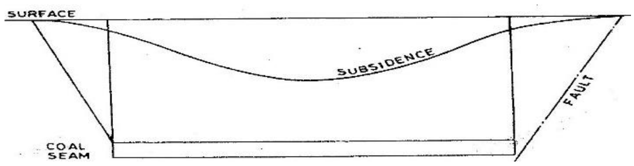
*Answer all the questions and each question carries 2 marks*

**29.** **(a)** Differentiate between: discontinuous and continuous subsidence. **(b)** Mention the factors affecting **(a)** subsidence **(b)** angle of draw.

**Discontinuous:** Subsidence is termed as discontinuous when large surface displacements over limited surface area occur due to shear action, which forms steps or discontinuities in the surface profile. It may develop suddenly or progressively, and may occur on a range of scales. Some of the types of discontinuous subsidence are shown in the Figure.




**Continuous:** It involves the formation of a smooth surface subsidence profile that is free of step changes. This type of subsidence is usually associated with the extraction of thin horizontal or flat-dipping orebodies overlain by weak non-brittle sedimentary strata mined by longwall method. This occurs in most longwall coal mining operations and in the metalliferous longwall mining at depth



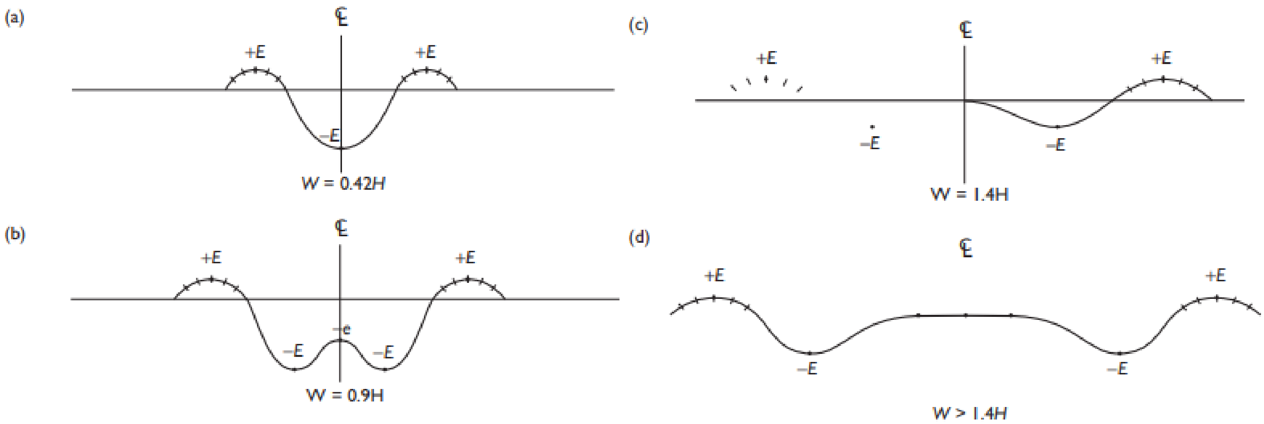

**Factors affecting (b) angle of draw.**

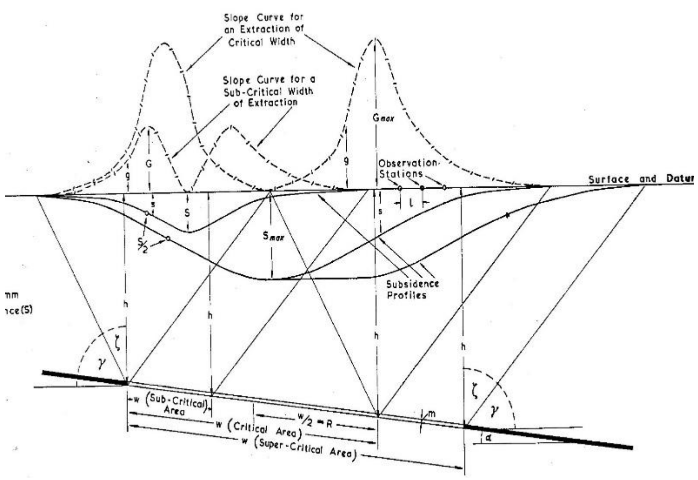
- depth,
- seam thickness, and
- local geology, especially major faults or fracture planes or self supporting strata above the coal seam.

**Factors affecting (a) subsidence**

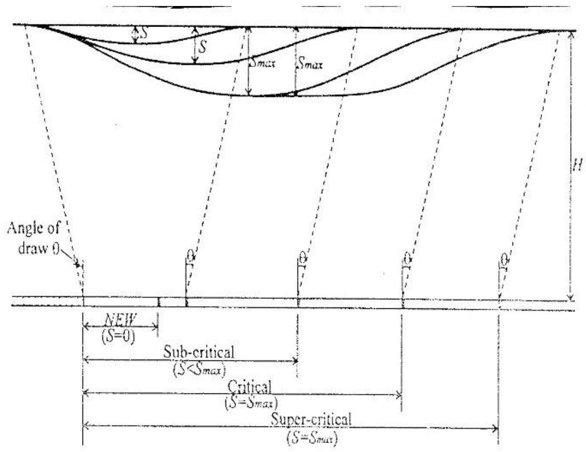
- Thickness of extracted materials
- Overlying mining areas
- Depth of mining
- Dip of mining deposit
- Competence and nature of mined and surrounding strata
- Near surface geology
- Geologic discontinuities
- Fractures and lineaments
- In-situ stresses
- Degree of extraction
- Surface topography
- Ground water (including water elevation and fluctuation)
- Mine area
- Method of mining
- Rate of advance
- Backfilling
- Time

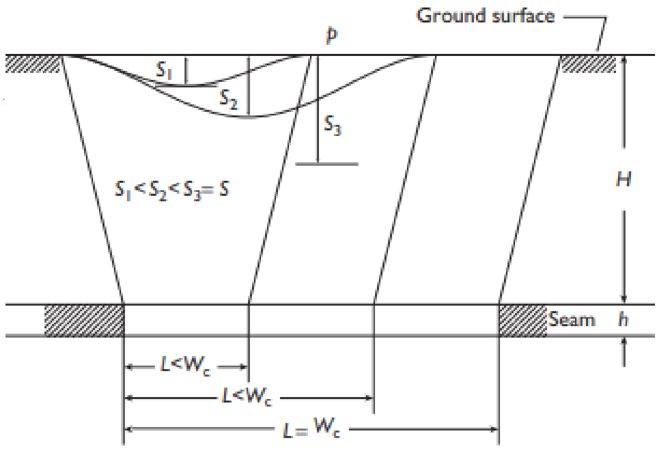
**30.** Draw strain profiles for **(a)** non-effective width **(b)** sub-critical width **(c)** critical width **(d)** super-critical width



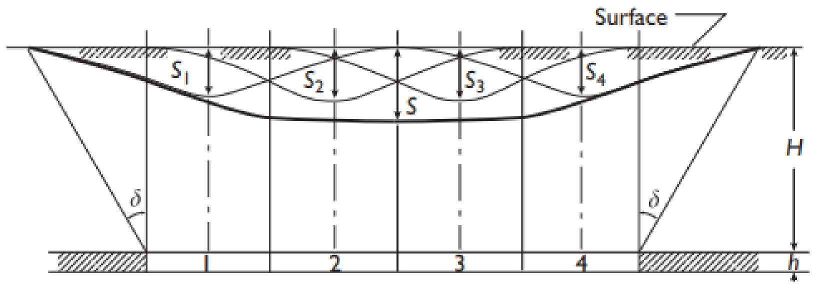



**31.** Draw subsidence profiles for **(a)** non-effective width **(b)** sub-critical width **(c)** critical width **(d)** super-critical width




**32.** Draw subsidence trough for **(a)** inclined coal seam **(b)** panels side by side with superposed subsidence profiles and **(c)** multi-seam superposition of subsidence profiles.



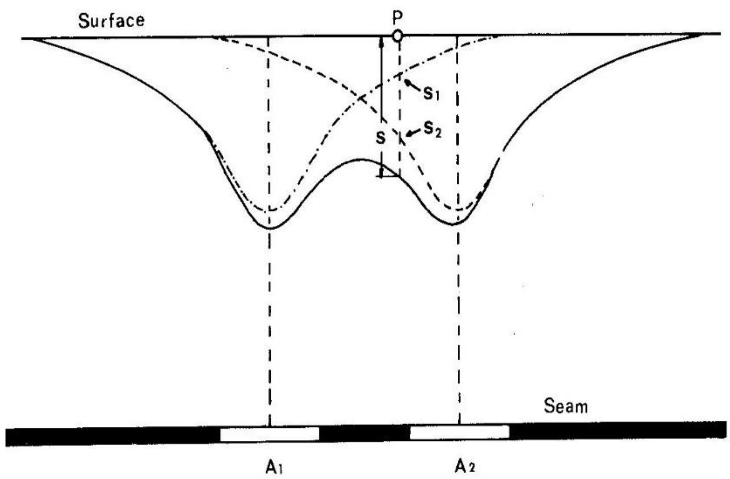
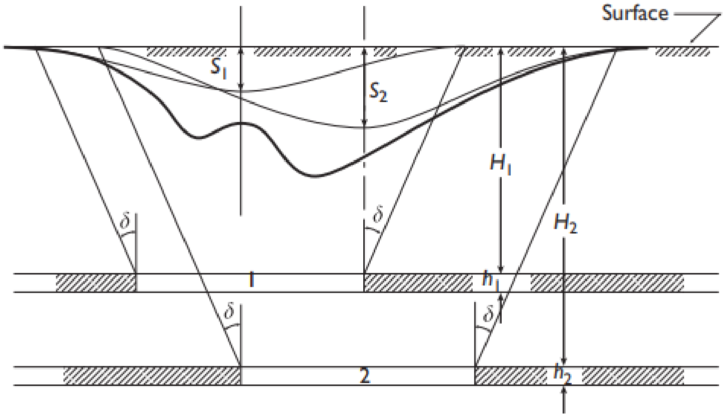
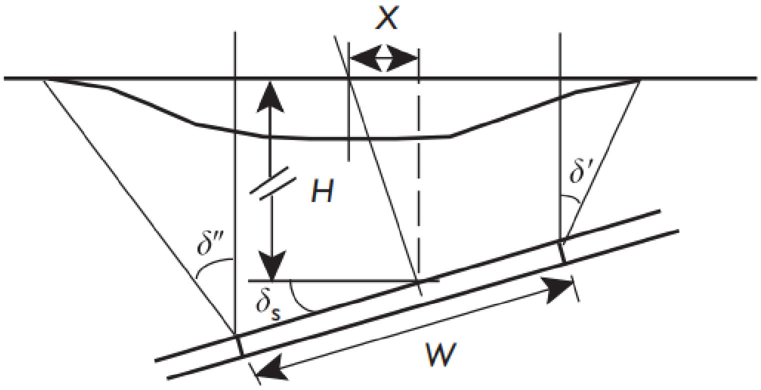
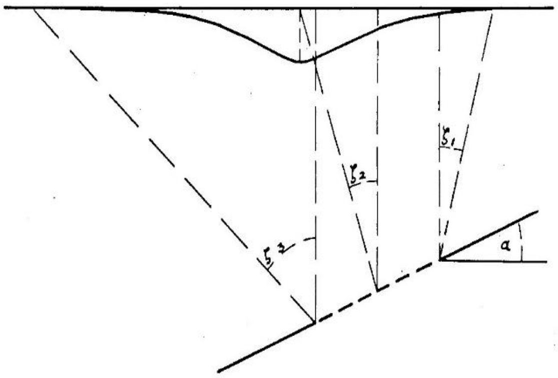



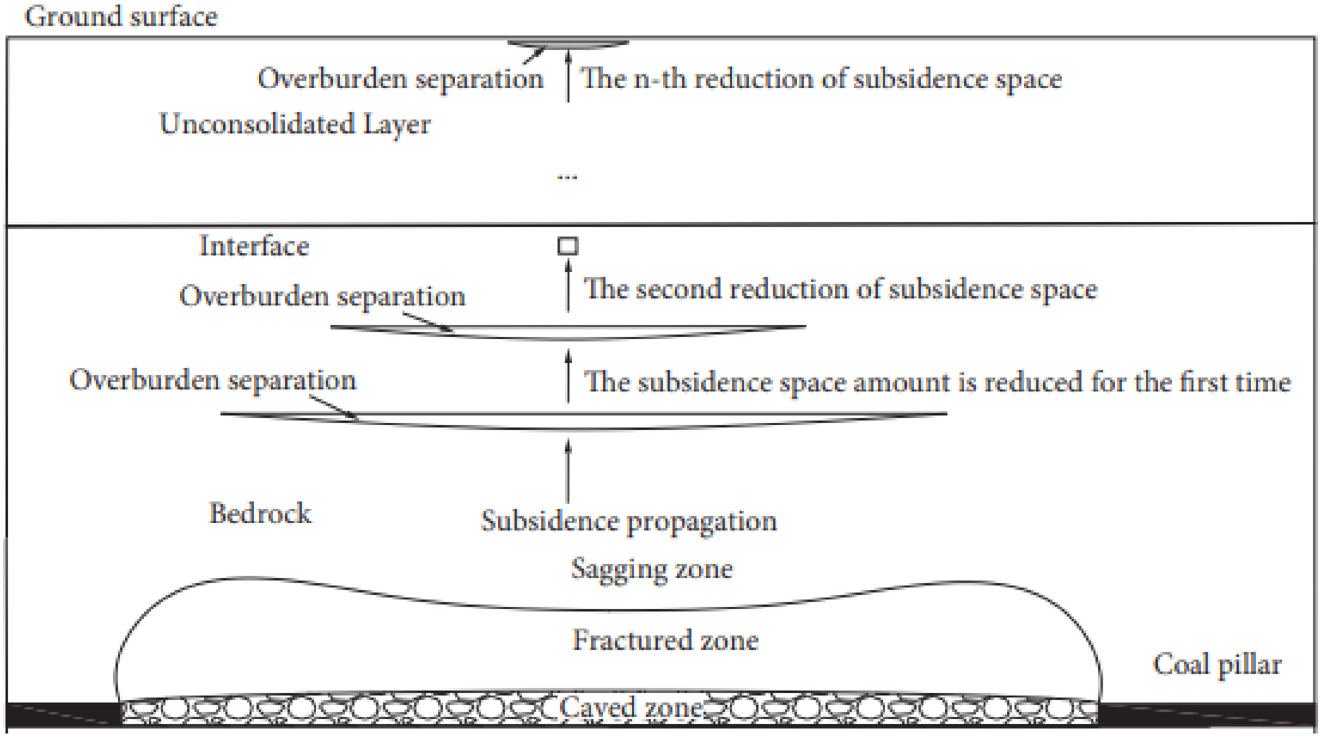

**Fig. 9.4.4** Superposition of surface subsidences (20).



**33.** Discuss subsidence at great depths (> 1000 m).




**34.** Discuss the impacts of mine subsidence and their control and prevention techniques.

**Impacts of mine subsidence**

- **Loss of water in surface water bodies (pond, river, nallah, jores etc)**
- **Depletion of water retention capability of sub-surface water table**
- **Depletion of water retention capability of aquifers**
- **Damage of buildings on surface**
- **Damage of rope ways**
- **Damage of high tension pylons**

Damage of underground cables
Damage of underground pipe lines
Damage of overlying virgin seams
Damage of overlying workings
Leakage of air, water and fire into underground working
Pollution of surface atmosphere due to harmful gasses from underground to surface
Depletion of water retention capability of sub-soil
Reduction in agricultural yield
Damage of forest

**Control and Prevention Mine Subsidence**

Partial extraction
Goaf treatment
Harmonic extraction
Safety pillars
Rapid Mining

**35.** What are the challenges faced while designing **(a)** two circular shafts of different diameter and **(b)** rectangular/circular/elliptical shafts in a row?

When two circular holes or shafts of different diameters are excavated in the same neighborhood, one may suppose that the large hole will significantly influence stress concentration near the small hole, while the small hole will have little influence on stress about the big hole.

When the holes in a row are not circular, two orientation questions arise: (1) the best orientation for the individual holes and (2) the best orientation for the row of holes. The first question may be tentatively answered by applying a rule of thumb for isolated, single openings that states the best orientation is to align the long axis of the hole with the direction of the major principal compressive stress. Whether this applies for multiple noncircular holes in a row when interaction occurs is an open question. The second question calls into play the rule of thumb that states the most favorable row of holes orientation is with the row axis, the line of centers, also parallel to the major compression. Whether this rule applies to noncircular holes is also an open question. Analytical or closed form solution to this problem is not known. However, a purely numerical approach allows for quantitative evaluation.

**36.** Discuss the procedure of installation of cable bolts.

**Procedures for installation of cable bolting**

**1. Selection of site for drilling**

As per the site requirement, where cable bolt is to be installed, the site is made level by chipping and dressing. The site is chosen based on mine manger's support design.

**2. Drilling**

• Drilling can be done by jumbo drill or by hydraulic bolter or any high capacity hydraulic drill rigs.
• Wet drilling by 42 mm drill bit is preferred using connecting rods.
• Before drilling, drill rods to be marked to the required length.
• Vertical hole of length of 5.950 m will be drilled at the site.
• Diameter of hole should be 43 mm.
• Flush the hole after drilling for dust particle removal.

**3. Resin insertion**

• Before inserting the resin into the top of the hole, the hole should be cleared by using flexible connecting rods.
• Then insert the resin (36 mm*1000 mm) capsule into the top of the hole by using the flexible connecting rods.
• Ensure that resin retainer is there for holding the resin capsule at hole top.
• Ensure resin is reached at the top of the hole, by observing the marker for resin position on the flexible connecting rods.
• Remove flexible connecting rods after ensuring that resin is placed at the top of the hole.

**4. Cable bolt preparation**

• Before using the cable for grouting, it should be clear of slush or mud all along its length.
• By placing it on level ground, breather tube should be attached and tied with plaster around the cable bolt up to resin retainer of the cable bolt.

**• The breather tube should be at least 150 mm more than the length of the cable bolt for checking air bubbles during cement injection.**

**5. Cable bolt insertion**

**• After the preparation of cable bolt, now insert the cable bolt into the hole with bulbs are at the top, up to reaching the resin capsule in the hole.**

**• Care should be taken not to puncture the resin capsule during insertion of cable bolt.**

**6. Spinning and churning the cable bolt**

**• After the insertion of cable bolt into the hole, now keep the churning adapter over the bolter and insert the bottom end of the cable bolt into the churning adapter**

**• Now slowly push the cable bolt upwards into the hole by spinning and thrust the cable bolt for churning.**

**• Then hold the cable bolt until setting time of the resin capsules reached.**

**• Remove the churning adapter.**

**• Now the cable bolt is point anchored with resin.**

**7. Accessories insertion**

**• After completion of cable bolt with resin anchor, insert grout tube about 25 cm into hole and 25 cm outside the hole and thickly pack the bottom of hole along cable bolt, breather tube and grout tube with cotton.**

**• Before fixing the bearing plate the grout tube and the breather tube should be passed through the holes of the bearing plates and attaches bearing plate with barrel and wedges.**

**• Now bolt is ready for pre-tensioning operation.**

**8. Pre-tensioning the cable bolt**

**• Place the standard tensioner unit suitable to cable bolt bearing plate-barrel and wedge combination, on to the cable bolt.**

**• Release some of the hand pressure tensioner unit to engage jaws.**

**• Pull tensioner unit down slightly whilst supporting it to make sure jaws have gripped the cable bolt, then apply tension for 30t by pressing the operating levels.**

**• After the require load is reached, release the pressure from the hydraulic ram by applying pressure to the release valve by pressing the pump treadle.**

**• Once the cable bolt has been tensioned and ram pressure release, then remove the tensioner unit by pushing the tensioner up and at the same time pull the release card down to disengage the jaws from the cable bolt.**

**• Care should be taken that the hydraulic ram fully retracted before tensioning the next bolt.**

**9. Post grouting with cement injection**

**• After completion of the pre-tensioning of cable bolt, now cable is ready for bottom up cement grouting.**

**• For cement injection process, place the mixing chamber at the location of cable bolt. • Add required quantity of cement and water for grouting cable bolt in the mixing chamber**

**• For mixing the cement and water, connect the mixing chamber to the compressor unit.**

**• Then start rotator slowly in the mixing chamber. • Now the mixture is ready for injection into the hole.**

**• Place the cement injection pump into the mixture.**

**• Then connect cement injection pump to compressor hose.**

**• Now connect the cement injection pump delivery to grout tube which is inserted earlier into the hole through bearing plate.**

**• Now place the breather tube delivery into the water to observe escape of air from hole. • Hold the pump delivery firmly, and start the pump and inject the cement into the hole.**

**• The cement injection should be continued until bubbles stopped from the breather tube delivery that means hole is fully filled up with cement.**

**• Then, remove cement injection pump delivery and breather tube delivery from water jug.**

**• Now tied up the grout tube to avoid back flowing of cement grout from hole.**

**Thus, complete operation of installing the cable bolt will be done.**

**37.** Two seam longwall mining occurs in panels 280 m wide by 2600 m long. The top seam is at a depth of 300 m and is 3.8 m thick; the bottom seam is at a depth of 320 m and is 4.5 m thick. The angle of draw and subsidence factors are 28° and 0.68. **(a)** Estimate maximum subsidence after mining a top seam panel and **(b)** then a bottom seam panel directly beneath the top seam panel. **(c)** Determine width of the subsidence trough.

**associated equation estimates of maximum subsidence under UK conditions are**

$$\left(\frac{S}{h}\right) = 0.9\left\{1 - e^{\left[-\frac{1}{2}\left(\frac{280/300}{0.45}\right)\right]^2}\right\} = 0.795$$

$$\therefore\ S = 3.8 * 0.795 = 3.02\ m\ (top)$$

$$\left(\frac{S}{h}\right) = 0.9\left\{1 - e^{\left[-\frac{1}{2}\left(\frac{280/320}{0.45}\right)\right]^2}\right\} = 0.764$$

$$\therefore\ S = 4.5 * 0.764 = 3.44\ m\ (bottom)$$

**Superposing the maximums is permissible because the panels are aligned horizontally.**
**Thus, S(max) = 3.02 + 3.44 = 6.46 m under UK conditions.**
**Under the given condition $S(top) = 3.02 * \frac{0.68}{0.9} = 2.28\ m;\ S(bottom) = 3.44 * \frac{0.68}{0.9} = 2.60\ m;$**

$$S(max) = 6.46 * \frac{0.68}{0.9} = 4.48\ m;$$

**The relative subsidence data for each panel is obtained from Table 8.1 and interpolation.**
**From the formula**
**W'= W + 2H tan(δ) = 280 + (2)(320) tan(28) = 620.**

**38.** A large vertical shaft is planned for an underground hard rock mine. Laboratory tests on core from exploration drilling show that $C_o = 21500\,psi,\ T_o = 1530\,psi, E = 6.25 * 10^6\,psi, G = 2.5 * 10^6\,psi, \gamma = 144\,pcf$ while premining stress measurements can be fit to the formulas $S_E = 200 + 0.3h, S_N = 600 + 0.9h, S_V = 1.1h$ where E, N, V refer to compass coordinates (x = east, y = north, z = up), stress is in kPa, depth h is in m, and compression is positive. Premining shear stresses are nil relative to compass coordinates. Find: **(a)** the most preferable shape of shaft (elliptical, rectangular, or ovaloid) when the cross section is 4.5 m × 9.0 m and **(b)** the resulting factors of safety with respect to compression and tension at a depth of 854 m.

**Table 1:** Peak stress concentration factor comparison of elliptical and rectangular sections at M= 1/3 (use proper reasoning for $K_c$ and $K_t$ of ovaloid shape based on the data mentioned below).

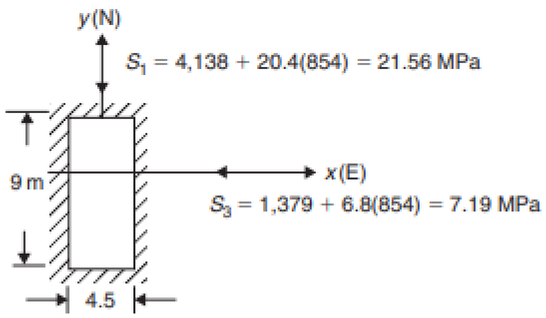
| Section | Case | | | |
|---|---|---|---|---|
| | M = 1/3, k = 1/2 | | M = 1/3, k = 2 | |
| | $K_c$ | $K_t$ | $K_c$ | $K_t$ |
| Ellipse | 1.67 | nil | 4.67 | -0.33 |
| Rectangle | 4.05 | -0.14 | 5.15 | -0.44 |

**Best orientation is with long axis parallel to $S_1$ in plan view**

$$S_x = 200 + 0.3h = 200 + 0.3 * 854 = 456.2\ kPa = 0.456\ MPa = \sigma_3 = 66.13\ psi$$
$$S_y = 600 + 0.9h = 1368.6\ kPa = 1.368\ MPa = \sigma_1 = 198\ psi$$
$$S_z = 1.1h = 939.4\ kPa = 0.939\ MPa = 136\ psi$$
$$S_y > S_x$$
$$S_y = \sigma_1 = 198\ psi$$
$$S_x = \sigma_3 = 66.13\ psi$$
$$k = \frac{1}{2} = \frac{W}{H} = 0.5$$
$$M = \frac{66.13}{198} = 0.33 = \frac{1}{3}$$

**Ellipse: $K_{max} = 1.67 = K_c$ and $K_{min} = nil = K_t$**
**Rectangle: $K_{max} = 4.05 = K_c$ and $K_{min} = -0.14 = K_t$**
**Ovaloid (rounded corners; less than rectangle K's)**
**∴ Ellipse is best, least $K_c, K_t$**
**$FS_t$ is not a factor**

$$FS_c = \frac{21500}{1.67 * 198} = 65.021$$



## *Section C: Numerical Type Questions – 48 marks*

*Answer all the questions and marks are assigned against each question*

**39.** A 3 ft thick circular concrete shaft liner is designed for an allowable stress of 3,500 psi in compression. Outside diameter is 24 ft. Determine the thickness of a steel liner with a safety factor of 1.5 that would support the same load but allow for a greater inside diameter and therefore a greater useful cross-sectional area of the shaft. UCS of steel liner is 30000 psi.

**(5 marks)**

**The load the concrete liner is designed to support is**

$$p_b = C_o\left[\frac{1-\left(\frac{a}{b}\right)^2}{2FS_c}\right] = 3500\left[\frac{1-\left(\frac{9}{12}\right)^2}{2}\right] = 766\ psi$$

**According to the first equation of (3.21)**

$$h = b\left[1-\sqrt{\left(\frac{C_o-2p_bFS}{C_o}\right)}\right] = 12*\left[1-\sqrt{\left(\frac{36000-2*766*1.5}{36000}\right)}\right] = 0.389\ ft = 4.67\ in$$

**The inside shaft diameter would increase from 21 ft to 23 ft.**
**Whether the increase is warranted would require a cost analysis.**
***Another Solution*:**

In case of Steel Liner; $FOS = 1.5 = \frac{Strength}{Stress} = \frac{36000}{Stress}$

**Stress on steel liner for 1.5 FOS is 24000 psi which is also maximum stress**

$$24000*\pi*[12^2-r^2] = 3500*\pi*[12^2-9^2]$$
$$144-r^2 = 9.1875$$
$$r = 11.61\ ft$$

**Thickness = 12-11.61 = 0.39 ft**

**40.** A mine in a flat seam at a depth of 520 m excavates panels with a face width of 260 m and a length of 2500 m. Mining is full seam height at 4.2 m. Compare **(a)** strain and **(b)** subsidence profiles when the angle of draw and subsidence factor are 37° and 0.9, respectively, with the case when they are 27° and 0.67. **(c)** In particular, locate the peak tensile and compressive strains in relation to the trough centre in both the cases.

**(9 + 9 + 4 = 22 marks)**

**: The width to depth ratio W/H = 260/520 = 0.5, so from the equation associated with Figure 8.15,**

$$\frac{S}{h} = a\left\{1-e^{\left[-\frac{1}{2}\left(\frac{W/H}{a/2}\right)\right]^2}\right\}$$

**where the constant a is the subsidence factor, that is, $a = \left(\frac{S}{h}\right)_{max} = S_f$, which is 0.9 in the UK but generally depends on the coal basin stratigraphic sequence.**

$$\frac{S}{h} = 0.9\left\{1-e^{\left[-\frac{1}{2}\left(\frac{0.50}{0.45}\right)\right]^2}\right\} = 0.415$$

$$S = 0.415*4.2 = 1.74\ m$$

**that would be the maximum subsidence over a single panel excavated under UK conditions.**
**Under the given conditions**

$$s' = s\left(\frac{S_f'}{S_f}\right) = 1.74*\left(\frac{0.67}{0.90}\right) = 1.30\ m$$

**From Table**

| $\frac{s}{S}$ | 0.00 | 0.05 | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 | 0.60 | 0.70 | 0.80 | 0.90 | 0.95 | 1.00 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\frac{x}{H}$ | 0.95 | 0.59 | 0.47 | 0.34 | 0.28 | 0.24 | 0.21 | 0.17 | 0.15 | 0.12 | 0.08 | 0.06 | 0 |

| s | 0.00 | 0.09 | 0.17 | 0.35 | 0.52 | 0.70 | 0.87 | 1.04 | 1.22 | 1.40 | 1.57 | 1.65 | 1.74 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| x | 494 | 306.8 | 244.4 | 176.8 | 145.6 | 124.8 | 109.2 | 88.4 | 78 | 62.4 | 41.6 | 31.2 | 0 |

Thus,

For x = 0.95 * 520 we have $x' = x\left(\frac{W'}{W_o}\right) = (0.95 * 520)\left[\frac{260+2*520\tan(27)}{260+2*520\tan(37)}\right] = 494\ m$

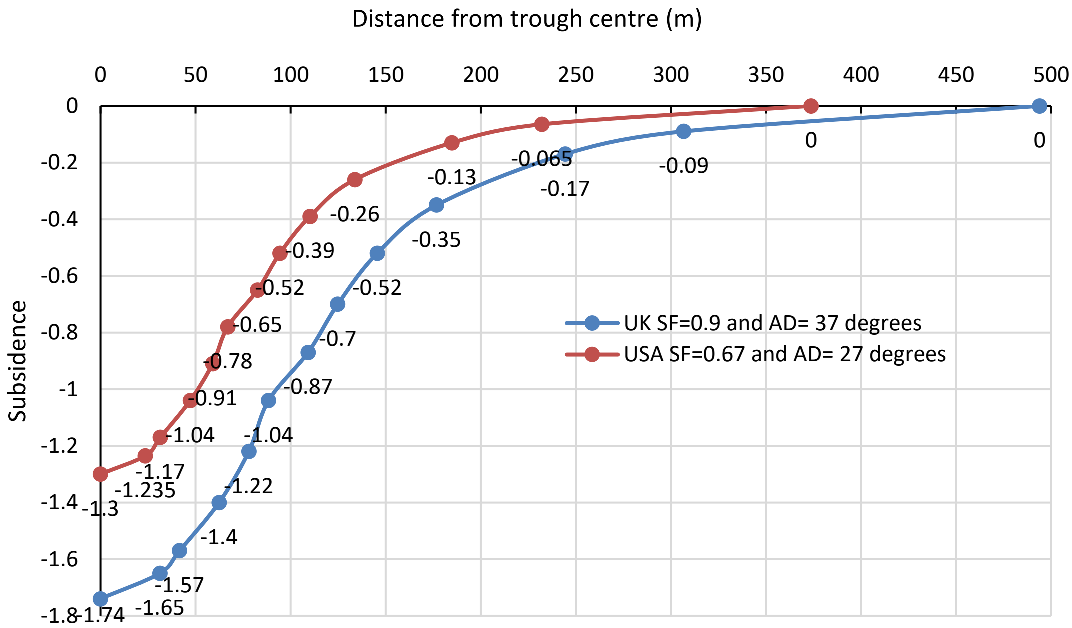
Similarly for x = 0.59*520 we have $x' = (0.59 * 520)\left[\frac{260+2*520\tan(27)}{260+2*520\tan(37)}\right] = 306.8\ m$

| $s_m = \frac{s}{S} * 1.30$ | 0.00 | 0.065 | 0.13 | 0.26 | 0.39 | 0.52 | 0.65 | 0.78 | 0.91 | 1.04 | 1.17 | 1.235 | 1.3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $x_m = \frac{x}{H} * 520 * x\left[\frac{W+2H\tan(\delta')}{W+2H\tan(\delta)}\right]$ | 373.8 | 232.2 | 184.9 | 133.8 | 110.2 | 94.4 | 82.6 | 66.9 | 59.0 | 47.2 | 31.5 | 23.6 | 0 |

after adjustments for the angle of draw and subsidence factor, Equations (8.37) and (8.38), are made.
The half-subsidence point occurs where s/S = 0.5, that is, where S = 0.65 m and x = 87.3 m from the panel center.
Trough width is

$W' = W + 2H\tan(\delta) = 260 + 2 * 520 * \tan(27) = 790\ m$

which agrees with the data in the table where zero subsidence occurs at 395 m from the trough center, so the trough width is 790 m.
A plot of the data for the subsidence profile is shown in the sketch.




The given angle of draw and subsidence factor correspond to UK conditions, so data from Tables 8.1, 8.2, and 8.3 can be used directly without adjustment.
The width to depth ratio is 0.5, so from Example 8.24, maximum subsidence is 1.74 m; the subsidence depth ratio (S/H) is then 1.74/520 = 3.346 x 10−3.
According to Table 8.3, E+ = 0.80 and E− = 1.35.
Hence the peak tensile and compressive strains are

$$E_t = (E+)\left(\frac{S}{H}\right) = (0.80)\left(\frac{1.74}{520}\right) = 0.80 * 3.346 * 10^{-3} = 2.677 * 10^{-3}$$

$$E_c = (E-)\left(\frac{S}{H}\right) = (1.35)\left(\frac{1.74}{520}\right) = 1.35 * 3.346 * 10^{-3} = 4.517 * 10^{-3}$$

From Figure 8.21, there is indeed a secondary compressive strain peak $e_c$ at the trough center.

The magnitude of this peak may be obtained from $\frac{-e}{-E} = e^{\left[-4\left(\frac{W}{H}-0.5\right)^2\right]} = e^{\left[-4(0.5-0.5)^2\right]} = 1$

$\therefore e_c = 4.517 * 10^{-3}$ which is compressive.

| | Extension (+E) | | | | | | | | Compression (-E) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $e/E$ | 0.0 | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 | 0.8 | 0 | 0.2 | 0.4 | 0.6 | 0.8 | 1.00 | 0.80 | 0.60 | 0.40 | 0.20 | 0.00 |
| $W/H$ | 0.96 | 0.60 | 0.51 | 0.43 | 0.38 | 0.32 | 0.27 | 0.21 | 0.18 | 0.16 | 0.12 | 0.09 | 0.02 | 0.00 | 0 | 0 | 0 | 0 |
| e x $10^{-3}$ | 0.0 | 0.53 | 1.07 | 1.61 | 2.14 | 2.67 | 2.14 | 0 | 0.90 | 1.81 | 2.71 | 3.61 | 4.52 | 3.61 | 2.71 | 1.81 | 0.90 | 0 |
| x | 499.2 | 312 | 265.2 | 223.6 | 197.6 | 166.4 | 140.4 | 109.2 | 93.6 | 83.2 | 62.4 | 46.8 | 10.4 | 0 | 0 | 0 | 0 | 0 |

These are the conditions of Example 8.27, although with a different angle of draw and subsidence factor. The width to depth ratio is again 0.5 for a single panel and from Example 8.24, the maximum subsidence is 1.30 m. From Table 8.3, E+ = 0.80 and E− = 1.35. Peak tensile and compressive strains are then

$$E_t = (E+)\left(\frac{S}{H}\right) = 0.80 * \frac{1.30}{520} = 2.000 * 10^{-3}$$

$$E_c = (E-)\left(\frac{S}{H}\right) = 1.35 * \frac{1.30}{520} = 3.375 * 10^{-3}$$

These strains are somewhat less than those in Example 8.27 for UK conditions.

The main reason is the lesser subsidence. Indeed, if no subsidence occurred, then no strains would be induced. From Figure 8.21, there is indeed a secondary compressive strain peak $e_c$ at the trough center. The magnitude of this peak may be obtained from the ratio

$\frac{-e}{-E} = e^{\left[-4\left(\frac{W}{H}-0.5\right)^2\right]} = e^{\left[-4(0.5-0.5)^2\right]} = 1 \therefore e_c = 4.517 * 10^{-3}$ which is compressive.

These are the conditions in Examples 8.27 and 8.29 (UK and US conditions, respectively). In the first case (UK case), the relative strains from Table 8.2 are

For x = 0.96 * 520 we have $x' = x\left(\frac{W'}{W_o}\right) = (0.96 * 520)\left[\frac{260+2*520\tan(27)}{260+2*520\tan(37)}\right] = 377.8\ m$

| | Extension (+E) | | | | | | | | Compression (-E) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $e'/E$ | 0.0 | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 | 0.8 | 0 | 0.2 | 0.4 | 0.6 | 0.8 | 1.00 | 0.80 | 0.60 | 0.40 | 0.20 | 0.00 |
| $x/H$ | 0.96 | 0.60 | 0.51 | 0.43 | 0.38 | 0.32 | 0.27 | 0.21 | 0.18 | 0.16 | 0.12 | 0.09 | 0.02 | 0.00 | 0 | 0 | 0 | 0 |
| e' x $10^{-3}$ | 0.0 | 0.4 | 0.8 | 1.2 | 1.6 | 2.0 | 1.6 | 0 | 0.67 | 1.35 | 2.02 | 2.70 | 3.37 | 2.70 | 2.02 | 1.35 | 0.67 | 0 |
| x' | 377.8 | 236.1 | 200.7 | 169.2 | 149.5 | 125.9 | 106.2 | 82.6 | 70.8 | 62.9 | 47.2 | 35.4 | 7.87 | 0 | 0 | 0 | 0 | 0 |

where the last row is obtained by linear interpolation.

Inspection of the compressive strain data shows the peak compression is reached very close to the trough center (x/H = 0.02).

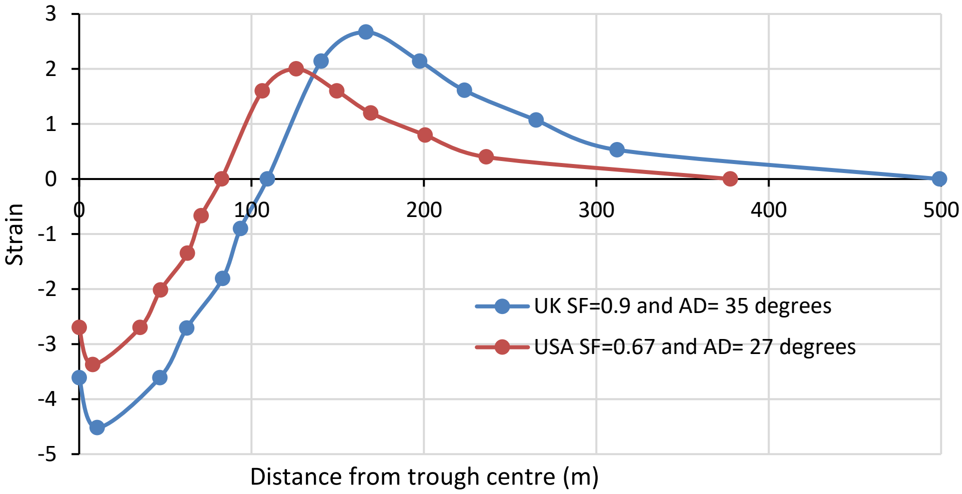
Distance entries after that point are zero indicating a sudden decrease of strain from near peak to zero at the trough center. However, this interpretation of the table data is incorrect.

In fact, the strain at the trough center is only slightly less than the peak compression. Indeed, from Figure 8.21, one sees that a secondary peak of compression occurs at the trough center that is only slightly less than the peak strain. In fact, the strain at the trough center is only slightly less than the peak compression. Indeed, from Figure 8.21, one sees that a secondary peak of compression occurs at the trough center that is only slightly less than the peak strain.

The actual strain and distance data are

where the strains should be divided by 1,000 and the distance is in meters. Again, tension is positive and compression is negative. In the US case, the data are

where the strains should be divided by 1,000 and the distance is in meters. The peak strain data and locations are summarized in the table.




**41.** Surface subsidence is observed over an area of 2.25 x $10^4$ $ft^2$ above an excavation 1,500 ft deep. Original excavation height is 15 ft and surface subsidence occurred eventually reaching 30 ft. There are 12 $ft^3$/ton of material. Estimate the bulking porosity and the tonnage of material removed after the original opening is excavated

$B = \frac{h}{(H+h)} = \frac{15}{(1500+15)} = 0.0099$

$$B = \frac{(h' - S)}{(H + h' - S)}$$

**where h' is a fictitious opening height associated with the given surface subsidence, that is h' void volume created by the initial excavation and the additional removal of solid. In this case**

$$h' = S + H\left(\frac{B}{1-B}\right) = 30 + 1500\left(\frac{0.0099}{1-0.0099}\right) = 45\ ft$$

**The tonnage of solid material removed is tons =** $\frac{\text{V (solids)}}{12} = \frac{(h'-h)A}{12} = \frac{(45-15)*2.25*10^4}{12} = 5.625 * 10^4$

**The tonnage of solid material removed is tons =** $\frac{30 * 2.25 * 10^4}{12} = 5.625 * 10^4$

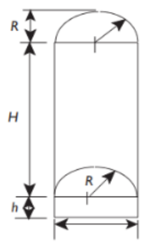
**OR**

Suppose an initial excavation is a cylinder 15 ft high with a 64 ft diameter. A hemispherical fall of rock occurs. Muck is removed and chimney formation begins. The hemispherical shape of back is maintained during caving that proceeds for 482 ft (back to top of chimney). No surface subsidence occurs. What bulking porosity is indicated?

**(5 marks)**

$$B = \frac{V_v}{V} = \frac{V_v}{V_s + V_v} = \frac{\pi R^2 h + \frac{2}{3}\pi * R^3}{\pi R^2 H + \pi R^2 h + \frac{2}{3}\pi * R^3} = \frac{h + \frac{2}{3}R}{H + h + \frac{2}{3}R}$$

$$= \frac{15 + \frac{2}{3} * 32}{(482 - 32) + 15 + \frac{2}{3} * 32} = 0.0747$$




**42.** A rectangular shaft 10 ft wide by 20 ft long is sunk vertically in ground where the state of premining stress is $\sigma_{xx} = 1141\ psi, \sigma_{yy} = 2059\ psi, \sigma_{zz} = 1600\ psi, \tau_{xy} = 221\ psi,\ \tau_{yz} = 0,\ \tau_{zx} = 0$ with compression positive, x = east, y = north, and z = up, in psi. Rock properties are: $C_o = 15000\ psi$, $T_o = 900\ psi, E = 4.5 * 10^6\ psi$: Find **(a)** the most favorable orientation of the shaft and **(b)** the safety factors in this orientation **(c)** Indicate by sketch where the peak stresses occur in cross section.

**(2 + 2 + 2 = 6 marks)**

$$\left.\begin{matrix} S_1 \\ S_3 \end{matrix}\right\} = \frac{S_{xx}+S_{yy}}{2} \pm \left[\left(\frac{S_{xx}-S_{yy}}{2}\right)^2 + \left(T_{xy}\right)^2\right]^{1/2}$$

$$\left.\begin{matrix} S_1 \\ S_3 \end{matrix}\right\} = \frac{1141+2059}{2} \pm \left[\left(\frac{1141-2059}{2}\right)^2 + (221)^2\right]^{1/2}$$

$$= 1600 \pm 509$$

$$S_1 = 2109\ psi$$

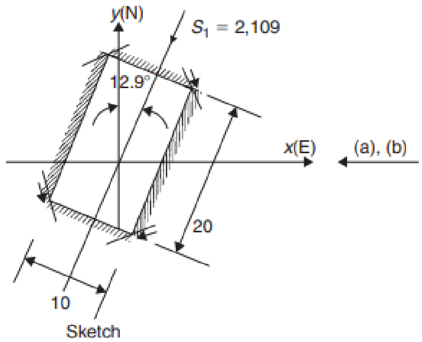
$$S_3 = 1091\ psi$$

$$\tan(2\beta) = \left[\frac{T_{xy}}{(S_{xx}-S_{yy})/2}\right] = \frac{221}{(1141-2509)/2} = -0.4815$$

$\beta = -12.9°$ (cw from y axis)

$$M = \frac{1091}{2109} = 0.517 \sim \frac{1}{2}$$

$$k = \frac{10}{20} = \frac{W}{H} = 0.5$$




Peak stress concentration occurs at corners

No tension, $FS_t$ is not a factor

$$FS_c = \frac{15000}{4.57 * 2109} = 1.56$$

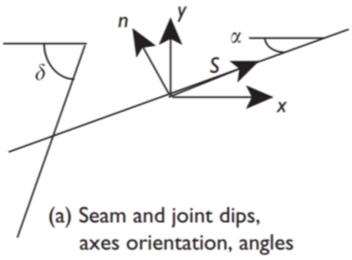
| k / k′ | M = 0 | | 1/5 | | 1/3 | | 1/2 | | 3/4 | | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | $K_{max}$ | $K_{min}$ | $K_{max}$ | $K_{min}$ | $K_{max}$ | $K_{min}$ | $K_{max}$ | $K_{min}$ | $K_{max}$ | $K_{min}$ | $K_{max}$ $K_{min}$ |
| 1/8 | 3.70 | –0.84 | 5.30 | –0.18 | 6.62 | 0.20 | 8.28 | 0.64 | 11.1 | 0.42 | 14.04 |
| 8 | 11.8 | –1.01 | 12.08 | –0.77 | 12.4 | –0.60 | 12.8 | –0.40 | 13.4 | –0.10 | 0.18 |
| 1/4 | 2.96 | –0.81 | 3.72 | –0.29 | 4.36 | 0.03 | 5.28 | 0.42 | 6.80 | 0.56 | 8.43 |
| 4 | 6.96 | –0.96 | 7.21 | –0.71 | 7.37 | –0.53 | 7.58 | –0.32 | 7.99 | 0.00 | 0.32 |
| 1/2 | 3.28 | –0.82 | 3.70 | –0.41 | 4.05 | –0.14 | 4.57 | 0.19 | 5.48 | 0.69 | 6.46 |
| 2 | 4.72 | –0.92 | 4.97 | –0.64 | 5.15 | –0.44 | 5.44 | –0.20 | 5.92 | 0.16 | 0.52 |
| 1 | 3.58 | –0.85 | 3.83 | –0.52 | 4.06 | –0.29 | 4.39 | –0.02 | 4.97 | 0.40 | 5.65 |
| | | | | | | | | | | | 0.80 |

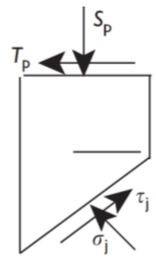
**43.** A room and pillar operation are planned at a depth of 964 ft in a seam dipping 13°. An extraction percentage of 50% is anticipated. The ratio of pre-excavation horizontal to vertical stress is estimated to be one-fourth. A joint set is present that dips 60° in the same direction as the seam. Rock and joint properties are given in the table. Determine if there is a threat to pillar stability. (Use $\gamma$ = 1 psi/ft)

**Table 4: Rock and joint properties.**

| Property material | Cohesion (psi) | Friction angle (°) |
|---|---|---|
| Rock | 2300 | 35 |
| Joint | 23 | 25 |

**(10 marks)**




**A threat to pillar stability exists if either the pillar safety factor or joint safety factor is less than one. Pillar strength and joint properties are given, while pillar stress may be determined from extraction ratio formulas. Joint stresses may be determined from equilibrium requirements. With reference to the sketch, the pre-excavation normal and shear stresses are**

$$S_n = \frac{S_x+S_y}{2} - \frac{S_x-S_y}{2}\cos 2\alpha + T_{xy}\sin 2\alpha = \frac{S_v(k+1)}{2} - \frac{S_v(k-1)}{2}\cos 26 + 0\sin 26$$

$$= \frac{964\left(\frac{1}{4}+1\right)}{2} - \frac{964\left(\frac{1}{4}-1\right)}{2}\cos 26 + 0\sin 26 = 927 \text{ psi}$$

$$T_s = -\left(\frac{S_x-S_y}{2}\right)\sin 2\alpha + T_{yz}\cos 2\alpha = -\frac{964\left(\frac{1}{4}-1\right)}{2}\sin 26 + 0\cos 26 = 158 \text{ psi}$$

**The pillar stresses are** $S_p = \frac{S_v}{1-R} = \frac{927}{1-\frac{1}{2}} = 1854$ **psi and** $T_p = \frac{T_s}{1-R} = \frac{158}{1-\frac{1}{2}} = 316$ **psi**

**The pillar factor of safety is given by** $FS = \frac{\frac{2c\cos\phi}{1-\sin(\phi-\beta)}}{\sqrt{(S_p)^2+(T_p)^2}} = \frac{\frac{2*2300\cos 35}{1-\sin(35-9.7)}}{\sqrt{(1854)^2+(316)^2}} = 3.5$

**Thus, the pillar safety factor indicates no threat of failure. The joint stresses are**

$$\sigma_j = \frac{S_p}{2}[1+\cos(2(\delta-\alpha))] - \frac{T_p}{2}\sin(2(\delta-\alpha)) = \frac{1854}{2}[1+\cos 94] - \frac{316}{2}\sin 94 = 705\ psi$$

$$\tau_j = \frac{1854}{2}\sin 94 - \frac{T_p}{2}[1+\cos(94)] = 1072 \text{ psi}$$

$$FS_j = \frac{\sigma_j \tan(\phi_j) + c_j}{\tau_j} = \frac{705\tan(25)+23}{1072} = 0.328$$

**Thus, joint failure poses a threat to pillar stability in this excavation plan.**

**OR**

Bord and Pillar Mining Method using Continuous Miner Technology is being implemented for extraction of 6 m thick coal seam at a depth of cover of 300 m with Rock Mass Rating (RMR) of 50 and Young's Modulus of roof of 3 GPa. The size of developed coal pillar is 30 m x 30 m and gallery width and height of 6 m and 4.5 m respectively. Use the appropriate empirical formulations to design the following geotechnical elements with appropriate diagram (wherever required)

**(i)** Cut-Out Distance (COD) **(0.5 mark)**
**(ii)** Extent of Pillar Spalling **(0.5 mark)**
**(iii)** Support design in galleries and junctions of 6.0 m (width) x 4.5 m (height) **(1 mark)**
**(iv)** Manner of Pillar Extraction based on COD **(2 marks)**
**(v)** Rib/Snook size/area with slice dimension **(2 marks)**
**(vi)** Support system in developed galleries **(2 marks)**
**(vii)** Location and grid pattern of roof bolts-based goaf edge support **(2 marks)**

**(i)** Cut-Out Distance (COD) **(0.5 mark)**

$$S = 14.61 + 1.98E - 2.12W = 14.61 + 1.98*3 - 2.12*6 = 7.83\ m \approx 8\ m$$

**(ii)** Extent of Pillar Spalling **(0.5 mark)**

$$x_0 = 0.00492hH = 0.00492*4.5*300 = 6.642\ m$$

**(iii)** Support design in galleries and junctions of 6.0 m (width) x 4.5 m (height) **(1 mark)**

$$Rock\ load\ in\ roadways = B\gamma(1.7 - 0.037RMR + 0.0002RMR^2)$$
$$= 6*2.5*(1.7 - 0.037*50 + 0.0002*50^2) = 5.25\ tons/m^2$$

**Let number of bolts = n; grouting strength is 20 tons for each bolt and grid pattern as 1.2 m x 1.2 m**

$$Support\ Resistance\ (S_r) = \frac{Number\ of\ support\ x\ Capacity\ of\ each\ support}{Area\ supported\ by\ one\ row\ of\ supports} = \frac{n * 20}{6 * 1.2}$$

$$Support\ safety\ factor = \frac{Support\ Resistance}{Rock\ Load} = \frac{\frac{n*20}{6*1.2}}{5.25} = 2$$

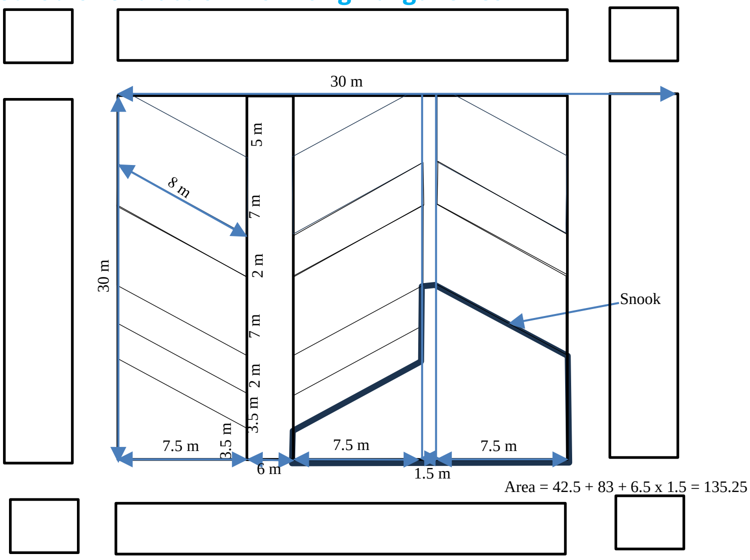
$$n = \frac{5.25 * 2 * 6 * 1.2}{20} = 3.78 \approx 4\ bolts$$

$$Rock\ load\ in\ junctions = 5B^{0.3}\gamma\left(1 - \frac{RMR}{100}\right)^2 = 5 * 6^{0.3} * 2.5\left(1 - \frac{50}{100}\right)^2 = 5.35\ tons/m^2$$

$$n = \frac{5.35 * 3 * 6 * 1}{20} \approx 5\ bolts$$

**(iv)** Manner of Pillar Extraction based on COD **(2 marks)**

**Since COD = 8m and pillar dimension is 30 m x 30 m, therefore, each fender width is 7.5 m. Combination of Christmas Tree and Split and slicing. Therefore, Christmas tree method extraction from split gallery of 6 m and split and slicing method of extraction from original galleries.**

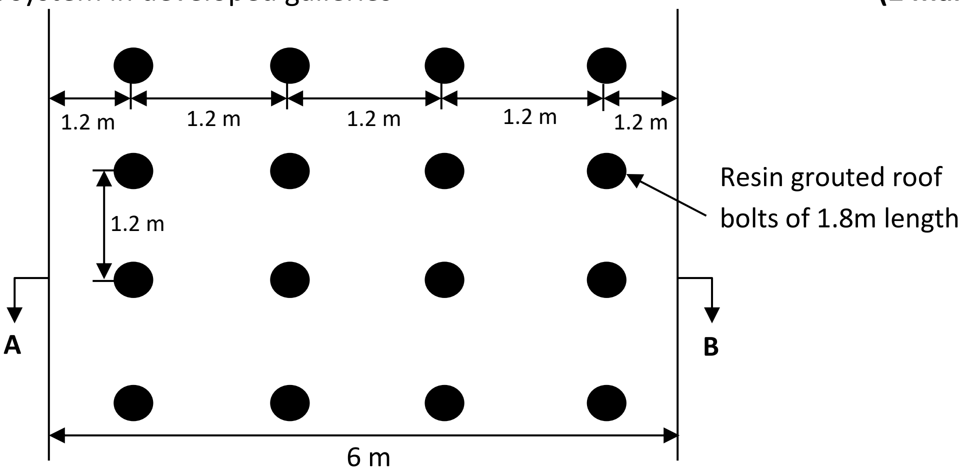


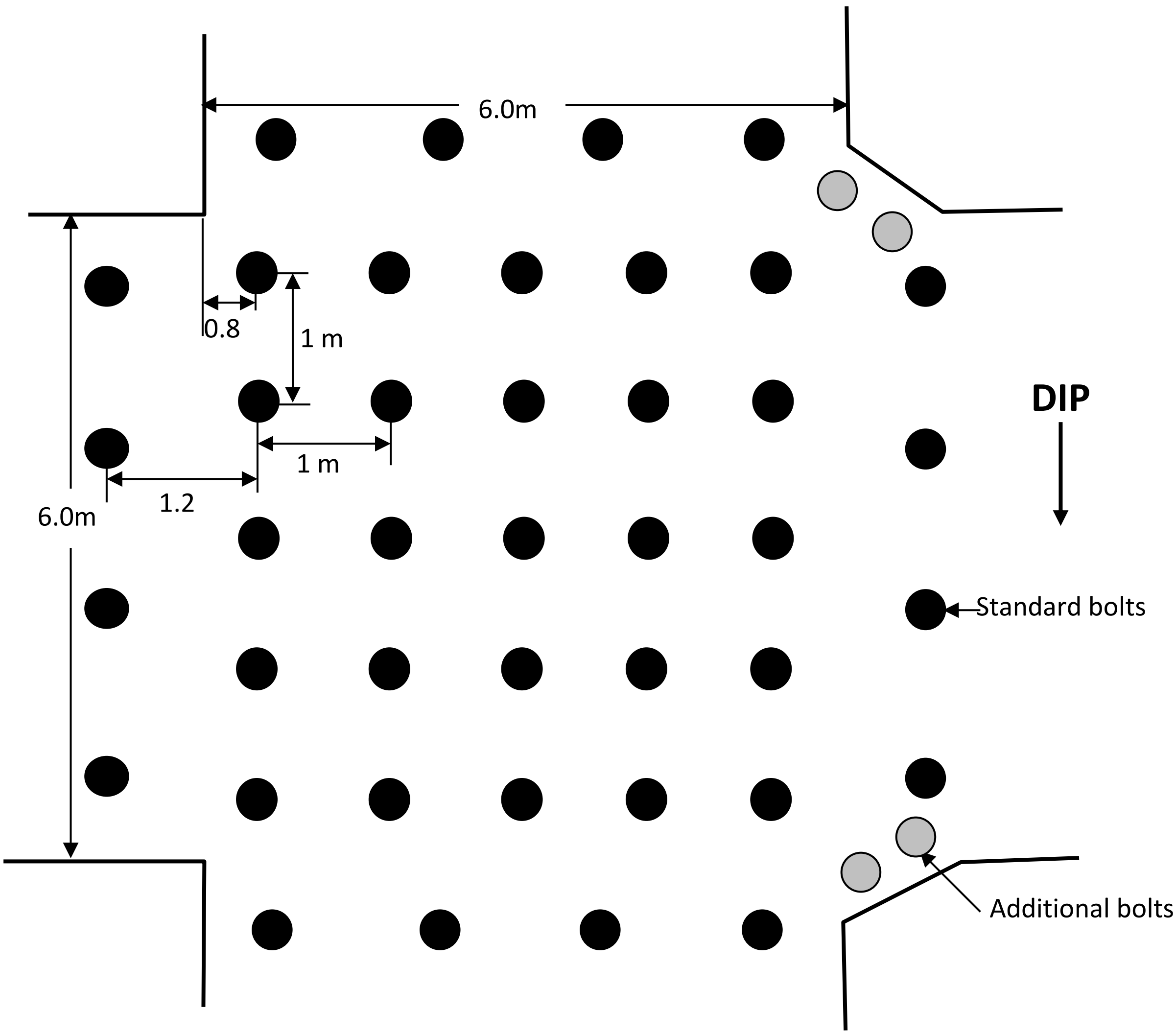

**(v)** Rib/Snook size/area with slice dimension **(2 marks)**

$$A = 0.16H^{0.89}R^{0.14}h^{0.31} = 0.16 * 300^{0.89} * 50^{0.14} * 4.5^{0.31} = 71\ m^2$$

$$w_e = 0.18H^{0.15}R^{0.10}h^{-0.03}A^{0.60} = 0.18 * 300^{0.15} * 50^{0.10} * 4.5^{-0.03} * 62^{0.60} = 7.7\ m$$

**(vi)** Support system in developed galleries **(2 marks)**






Proposed support system at junctions

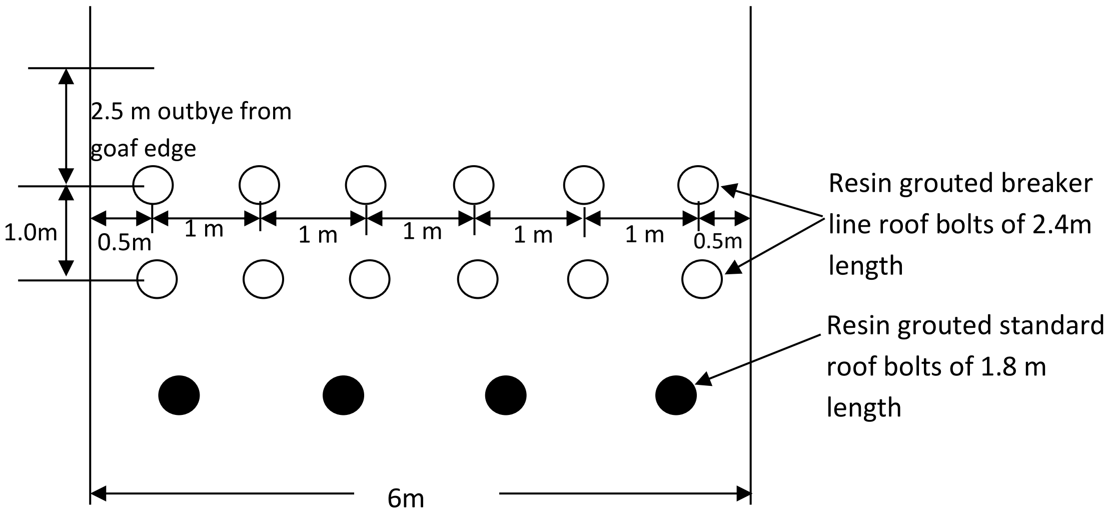
**(vii)** Location and grid pattern of roof bolts-based goaf edge support **(2 marks)**

**For 2 m out-bye from goaf edge: RLH = 115.22 $H^{0.12}$ $R^{-1.20}$= 115.22 *$300^{0.12}$ * $50^{-1.20}$= 2.1 m**

**Rock Load= 2.5 * 2.1 = 5.25 tons/$m^2$**

$$\text{Support safety factor} = \frac{\text{Support Resistance}}{\text{Rock Load}} = \frac{\frac{n*20}{6*0.8}}{5.25} = 4$$

$$n = \frac{5.25 * 4 * 6 * 1}{20} = 6.3 \approx 6 \text{ bolts}$$




Proposed breaker line support system to be installed

**Table 5: Formulae to be used (where symbols have their own meaning).**

<table>
<tr><td colspan="2">For Rib/Snook<br>$A = 0.16H^{0.89}R^{0.14}h^{0.31}$<br>$\frac{w_e}{h} = 0.18H^{0.15}R^{0.10}h^{-1.03}A^{0.60}$<br>$w_e = 0.18H^{0.15}R^{0.10}h^{-0.03}A^{0.60}$</td><td colspan="2">For RBBLS<br>For 0 m out-bye from goaf edge: RLH = 11.67 $H^{0.58}$ $R^{-1.14}$<br>For 1 m out-bye from goaf edge: RLH = 66.32 $H^{0.31}$ $R^{-1.26}$<br>For 2 m out-bye from goaf edge: RLH = 115.22 $H^{0.12}$ $R^{-1.20}$</td></tr>
<tr><td colspan="3">For Support system in developed galleries<br>Rock load in roadways = $B\gamma(1.7 - 0.037RMR + 0.0002RMR^2)$<br>$Rock\ load\ in\ junctions\ =\ 5B^{0.3}\gamma\left(1-\frac{RMR}{100}\right)^2$</td><td>For Pillar Spalling<br>$x_0 = 0.00492hH$<br>For Cut-out distance<br>$S = 14.61 + 1.98E - 2.12W$</td></tr>
<tr><td>$\frac{S}{h} = 0.9\left\{1 - e^{\left[-2.47\left(\frac{x}{H}\right)\right]^2}\right\}$</td><td colspan="2">$\frac{S'}{S} = \left\{1 - e^{\left[-2.70\left(\frac{x}{H}\right)\right]^2}\right\}$</td><td>$\frac{-e}{-E} = e^{\left[-4\left(\frac{W}{H}-0.5\right)^2\right]}$</td></tr>
<tr><td colspan="2">$p_b = C_o\left[\frac{1-\left(\frac{a}{b}\right)^2}{2FS_c}\right]$</td><td colspan="2">$h = b\left[1-\sqrt{\left(\frac{C_o - 2p_bFS}{C_o}\right)}\right]$</td></tr>
</table>

**Table 6: Stress concentration factors about a rectangular section.**

| k<br>k′ | M | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0 | | 1/5 | | 1/3 | | 1/2 | | 3/4 | | 1 |
| | $K_{max}$ | $K_{min}$ | $K_{max}$ | $K_{min}$ | $K_{max}$ | $K_{min}$ | $K_{max}$ | $K_{min}$ | $K_{max}$ | $K_{min}$ | $K_{max}$<br>$K_{min}$ |
| 1/8 | 3.70 | −0.84 | 5.30 | −0.18 | 6.62 | 0.20 | 8.28 | 0.64 | 11.1 | 0.42 | 14.04 |
| 8 | 11.8 | −1.01 | 12.08 | −0.77 | 12.4 | −0.60 | 12.8 | −0.40 | 13.4 | −0.10 | 0.18 |
| 1/4 | 2.96 | −0.81 | 3.72 | −0.29 | 4.36 | 0.03 | 5.28 | 0.42 | 6.80 | 0.56 | 8.43 |
| 4 | 6.96 | −0.96 | 7.21 | −0.71 | 7.37 | −0.53 | 7.58 | −0.32 | 7.99 | 0.00 | 0.32 |
| 1/2 | 3.28 | −0.82 | 3.70 | −0.41 | 4.05 | −0.14 | 4.57 | 0.19 | 5.48 | 0.69 | 6.46 |
| 2 | 4.72 | −0.92 | 4.97 | −0.64 | 5.15 | −0.44 | 5.44 | −0.20 | 5.92 | 0.16 | 0.52 |
| 1 | 3.58 | −0.85 | 3.83 | −0.52 | 4.06 | −0.29 | 4.39 | −0.02 | 4.97 | 0.40 | 5.65 |
| | | | | | | | | | | | 0.80 |

**Table 7: Subsidence profile points at various width to depth ratios (from SHB, 1975)**

| s/S W/H | 0.00 | 0.05 | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 X/H | 0.60 | 0.70 | 0.80 | 0.90 | 0.95 | 1.00 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.20 | 0.80 | 0.64 | 0.57 | 0.48 | 0.41 | 0.37 | 0.32 | 0.28 | 0.23 | 0.19 | 0.13 | 0.08 | 0.00 |
| 0.24 | 0.82 | 0.62 | 0.53 | 0.43 | 0.36 | 0.32 | 0.28 | 0.24 | 0.20 | 0.16 | 0.11 | 0.07 | 0.00 |
| 0.28 | 0.84 | 0.61 | 0.51 | 0.39 | 0.33 | 0.28 | 0.24 | 0.21 | 0.18 | 0.14 | 0.09 | 0.07 | 0.00 |
| 0.30 | 0.85 | 0.61 | 0.50 | 0.38 | 0.32 | 0.27 | 0.23 | 0.20 | 0.17 | 0.13 | 0.09 | 0.06 | 0.00 |
| 0.34 | 0.87 | 0.60 | 0.49 | 0.36 | 0.30 | 0.25 | 0.22 | 0.19 | 0.16 | 0.12 | 0.08 | 0.06 | 0.00 |
| 0.38 | 0.89 | 0.60 | 0.48 | 0.35 | 0.29 | 0.24 | 0.21 | 0.18 | 0.15 | 0.12 | 0.08 | 0.06 | 0.00 |
| 0.40 | 0.90 | 0.59 | 0.47 | 0.34 | 0.28 | 0.24 | 0.21 | 0.18 | 0.15 | 0.12 | 0.08 | 0.06 | 0.00 |
| 0.44 | 0.92 | 0.59 | 0.47 | 0.33 | 0.28 | 0.23 | 0.20 | 0.17 | 0.15 | 0.12 | 0.08 | 0.06 | 0.00 |
| 0.48 | 0.94 | 0.59 | 0.47 | 0.33 | 0.28 | 0.23 | 0.20 | 0.17 | 0.15 | 0.12 | 0.08 | 0.06 | 0.00 |
| 0.50 | 0.95 | 0.59 | 0.47 | 0.34 | 0.28 | 0.24 | 0.21 | 0.17 | 0.15 | 0.12 | 0.08 | 0.06 | 0.00 |
| 0.54 | 0.97 | 0.59 | 0.47 | 0.34 | 0.29 | 0.25 | 0.21 | 0.18 | 0.15 | 0.12 | 0.08 | 0.06 | 0.00 |
| 0.58 | 0.99 | 0.59 | 0.47 | 0.35 | 0.30 | 0.25 | 0.22 | 0.18 | 0.16 | 0.13 | 0.09 | 0.06 | 0.00 |
| 0.60 | 1.00 | 0.59 | 0.47 | 0.36 | 0.30 | 0.26 | 0.22 | 0.19 | 0.16 | 0.13 | 0.09 | 0.06 | 0.00 |
| 0.64 | 1.02 | 0.59 | 0.48 | 0.37 | 0.31 | 0.27 | 0.23 | 0.20 | 0.17 | 0.13 | 0.09 | 0.06 | 0.00 |
| 0.68 | 1.04 | 0.60 | 0.49 | 0.38 | 0.32 | 0.28 | 0.24 | 0.21 | 0.17 | 0.14 | 0.10 | 0.07 | 0.00 |
| 0.70 | 1.05 | 0.60 | 0.49 | 0.39 | 0.33 | 0.29 | 0.25 | 0.21 | 0.18 | 0.14 | 0.10 | 0.07 | 0.00 |
| 0.74 | 1.07 | 0.61 | 0.50 | 0.40 | 0.34 | 0.30 | 0.26 | 0.23 | 0.19 | 0.15 | 0.10 | 0.07 | 0.00 |
| 0.78 | 1.09 | 0.63 | 0.52 | 0.42 | 0.36 | 0.32 | 0.28 | 0.24 | 0.20 | 0.16 | 0.11 | 0.08 | 0.00 |
| 0.80 | 1.10 | 0.63 | 0.52 | 0.42 | 0.36 | 0.32 | 0.28 | 0.25 | 0.21 | 0.17 | 0.11 | 0.08 | 0.00 |
| 0.84 | 1.12 | 0.65 | 0.54 | 0.44 | 0.38 | 0.34 | 0.30 | 0.26 | 0.22 | 0.18 | 0.12 | 0.09 | 0.00 |
| 0.88 | 1.14 | 0.67 | 0.56 | 0.45 | 0.40 | 0.36 | 0.32 | 0.28 | 0.24 | 0.20 | 0.13 | 0.10 | 0.00 |
| 0.90 | 1.15 | 0.68 | 0.57 | 0.46 | 0.40 | 0.36 | 0.32 | 0.29 | 0.25 | 0.20 | 0.14 | 0.10 | 0.00 |
| 0.94 | 1.17 | 0.69 | 0.58 | 0.48 | 0.42 | 0.38 | 0.34 | 0.31 | 0.26 | 0.22 | 0.16 | 0.11 | 0.00 |
| 0.98 | 1.19 | 0.71 | 0.60 | 0.50 | 0.44 | 0.40 | 0.36 | 0.33 | 0.28 | 0.24 | 0.17 | 0.12 | 0.00 |
| 1.00 | 1.20 | 0.72 | 0.61 | 0.51 | 0.45 | 0.41 | 0.37 | 0.33 | 0.29 | 0.24 | 0.18 | 0.13 | 0.00 |
| 1.10 | 1.25 | 0.77 | 0.65 | 0.55 | 0.50 | 0.45 | 0.42 | 0.38 | 0.34 | 0.29 | 0.21 | 0.16 | 0.00 |
| 1.20 | 1.30 | 0.81 | 0.70 | 0.60 | 0.54 | 0.50 | 0.46 | 0.42 | 0.38 | 0.33 | 0.25 | 0.19 | 0.00 |
| 1.30 | 1.35 | 0.86 | 0.75 | 0.65 | 0.59 | 0.55 | 0.51 | 0.47 | 0.43 | 0.38 | 0.30 | 0.23 | 0.00 |
| 1.40 | 1.40 | 0.91 | 0.80 | 0.70 | 0.64 | 0.60 | 0.56 | 0.52 | 0.48 | 0.43 | 0.35 | 0.27 | 0.00 |
| 1.60 | 1.50 | 1.01 | 0.90 | 0.80 | 0.74 | 0.70 | 0.66 | 0.62 | 0.58 | 0.53 | 0.45 | 0.37 | 0.05 |
| 1.80 | 1.60 | 1.11 | 1.00 | 0.90 | 0.84 | 0.80 | 0.76 | 0.72 | 0.68 | 0.63 | 0.55 | 0.47 | 0.10 |
| 2.00 | 1.70 | 1.21 | 1.09 | 0.99 | 0.94 | 0.90 | 0.86 | 0.82 | 0.78 | 0.73 | 0.65 | 0.57 | 0.16 |
| 2.20 | 1.80 | 1.31 | 1.19 | 1.09 | 1.04 | 1.00 | 0.96 | 0.92 | 0.88 | 0.83 | 0.75 | 0.67 | 0.23 |
| 2.40 | 1.90 | 1.41 | 1.29 | 1.19 | 1.14 | 1.10 | 1.06 | 1.02 | 0.98 | 0.93 | 0.85 | 0.77 | 0.31 |
| 2.60 | 2.00 | 1.51 | 1.39 | 1.29 | 1.24 | 1.19 | 1.16 | 1.12 | 1.08 | 1.03 | 0.95 | 0.87 | 0.41 |

**Table 6: Peak strain multipliers.**

| W/H | E+ | E- |
|---|---|---|
| 0.10 | 0.25 | 2.40 |
| 0.20 | 0.52 | 2.25 |
| 0.30 | 0.70 | 2.00 |
| 0.40 | 0.77 | 1.73 |
| 0.50 | 0.80 | 1.35 |
| 0.60 | 0.75 | 1.06 |
| 0.70 | 0.68 | 0.86 |
| 0.80 | 0.65 | 0.70 |
| 0.90 | 0.65 | 0.60 |
| 1.00 | 0.65 | 0.55 |
| 1.10 | 0.65 | 0.54 |
| 1.20 | 0.65 | 0.53 |
| 1.30 | 0.65 | 0.52 |
| 1.40 | 0.65 | 0.51 |

Note
a Tension (+),compression (–).

**Table 8: Strain profile points at various width to depth ratios (from SHB, 1975)**

| e/E W/H | 0.00 | 0.20 | 0.40 | 0.60 | 0.80 | 1.00 | 0.80 | 0.00 | 0.20 X/H | 0.40 | 0.60 | 0.80 | 1.00 | 0.80 | 0.60 | 0.40 | 0.20 | 0.00 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.20 | 0.80 | 0.74 | 0.69 | 0.66 | 0.61 | 0.49 | 0.42 | 0.32 | 0.27 | 0.23 | 0.18 | 0.12 | 0.00 | 0.00 | 0 | 0 | 0 | 0 |
| 0.24 | 0.82 | 0.70 | 0.63 | 0.60 | 0.54 | 0.44 | 0.37 | 0.28 | 0.23 | 0.20 | 0.15 | 0.10 | 0.00 | 0.00 | 0 | 0 | 0 | 0 |
| 0.28 | 0.84 | 0.66 | 0.58 | 0.54 | 0.49 | 0.39 | 0.33 | 0.24 | 0.21 | 0.17 | 0.13 | 0.08 | 0.00 | 0.00 | 0 | 0 | 0 | 0 |
| 0.32 | 0.86 | 0.63 | 0.55 | 0.49 | 0.45 | 0.35 | 0.30 | 0.22 | 0.19 | 0.16 | 0.12 | 0.07 | 0.00 | 0.00 | 0 | 0 | 0 | 0 |
| 0.36 | 0.88 | 0.62 | 0.53 | 0.46 | 0.42 | 0.33 | 0.29 | 0.21 | 0.18 | 0.15 | 0.11 | 0.07 | 0.00 | 0.00 | 0 | 0 | 0 | 0 |
| 0.40 | 0.90 | 0.61 | 0.52 | 0.45 | 0.40 | 0.32 | 0.28 | 0.21 | 0.18 | 0.15 | 0.11 | 0.07 | 0.00 | 0.00 | 0 | 0 | 0 | 0 |
| 0.44 | 0.92 | 0.60 | 0.51 | 0.43 | 0.39 | 0.31 | 0.27 | 0.20 | 0.18 | 0.15 | 0.11 | 0.07 | 0.01 | 0.00 | 0 | 0 | 0 | 0 |
| 0.48 | 0.94 | 0.60 | 0.51 | 0.43 | 0.38 | 0.31 | 0.27 | 0.20 | 0.18 | 0.15 | 0.12 | 0.08 | 0.01 | 0.00 | 0 | 0 | 0 | 0 |
| 0.52 | 0.96 | 0.60 | 0.51 | 0.43 | 0.38 | 0.32 | 0.27 | 0.21 | 0.18 | 0.16 | 0.12 | 0.09 | 0.02 | 0.00 | 0 | 0 | 0 | 0 |
| 0.56 | 0.98 | 0.61 | 0.51 | 0.44 | 0.39 | 0.33 | 0.28 | 0.22 | 0.19 | 0.17 | 0.13 | 0.10 | 0.03 | 0.00 | 0 | 0 | 0 | 0 |
| 0.60 | 1.00 | 0.62 | 0.52 | 0.45 | 0.40 | 0.34 | 0.29 | 0.22 | 0.20 | 0.18 | 0.14 | 0.11 | 0.05 | 0.00 | 0 | 0 | 0 | 0 |
| 0.64 | 1.02 | 0.63 | 0.53 | 0.46 | 0.41 | 0.35 | 0.31 | 0.23 | 0.21 | 0.19 | 0.15 | 0.12 | 0.06 | 0.00 | 0 | 0 | 0 | 0 |
| 0.68 | 1.04 | 0.64 | 0.54 | 0.47 | 0.43 | 0.37 | 0.32 | 0.24 | 0.22 | 0.20 | 0.16 | 0.13 | 0.07 | 0.00 | 0 | 0 | 0 | 0 |
| 0.72 | 1.06 | 0.66 | 0.55 | 0.49 | 0.44 | 0.38 | 0.34 | 0.26 | 0.24 | 0.21 | 0.17 | 0.15 | 0.09 | 0.00 | 0 | 0 | 0 | 0 |
| 0.76 | 1.08 | 0.67 | 0.57 | 0.51 | 0.46 | 0.40 | 0.36 | 0.27 | 0.25 | 0.22 | 0.19 | 0.16 | 0.10 | 0.01 | 0 | 0 | 0 | 0 |
| 0.80 | 1.10 | 0.69 | 0.58 | 0.53 | 0.48 | 0.42 | 0.37 | 0.29 | 0.26 | 0.24 | 0.20 | 0.17 | 0.11 | 0.02 | 0 | 0 | 0 | 0 |
| 0.84 | 1.12 | 0.71 | 0.60 | 0.54 | 0.49 | 0.44 | 0.39 | 0.30 | 0.28 | 0.25 | 0.22 | 0.19 | 0.13 | 0.04 | 0 | 0 | 0 | 0 |
| 0.88 | 1.14 | 0.73 | 0.62 | 0.56 | 0.51 | 0.46 | 0.41 | 0.32 | 0.29 | 0.27 | 0.24 | 0.21 | 0.15 | 0.05 | 0.01 | 0 | 0 | 0 |
| 0.92 | 1.16 | 0.75 | 0.64 | 0.58 | 0.53 | 0.47 | 0.43 | 0.34 | 0.31 | 0.29 | 0.25 | 0.22 | 0.16 | 0.07 | 0.02 | 0 | 0 | 0 |
| 0.96 | 1.18 | 0.77 | 0.66 | 0.60 | 0.55 | 0.49 | 0.45 | 0.35 | 0.33 | 0.30 | 0.27 | 0.24 | 0.18 | 0.09 | 0.04 | 0 | 0 | 0 |
| 1.00 | 1.20 | 0.79 | 0.68 | 0.62 | 0.57 | 0.51 | 0.47 | 0.37 | 0.35 | 0.32 | 0.29 | 0.26 | 0.20 | 0.10 | 0.05 | 0 | 0 | 0 |
| 1.20 | 1.30 | 0.88 | 0.77 | 0.71 | 0.66 | 0.61 | 0.56 | 0.46 | 0.44 | 0.41 | 0.38 | 0.35 | 0.29 | 0.20 | 0.13 | 0.07 | 0.02 | 0 |
| 1.40 | 1.40 | 0.98 | 0.87 | 0.81 | 0.75 | 0.70 | 0.66 | 0.56 | 0.53 | 0.51 | 0.48 | 0.45 | 0.39 | 0.30 | 0.23 | 0.17 | 0.1 | 0 |
| 1.60 | 1.50 | 1.08 | 0.97 | 0.91 | 0.85 | 0.80 | 0.76 | 0.66 | 0.63 | 0.61 | 0.58 | 0.55 | 0.49 | 0.40 | 0.33 | 0.27 | 0.2 | 0.03 |
| 1.80 | 1.60 | 1.17 | 1.07 | 1.01 | 0.95 | 0.90 | 0.86 | 0.76 | 0.73 | 0.71 | 0.68 | 0.65 | 0.59 | 0.50 | 0.43 | 0.37 | 0.3 | 0.1 |
| 2.00 | 1.70 | 1.28 | 1.17 | 1.11 | 1.05 | 1.00 | 0.96 | 0.86 | 0.84 | 0.81 | 0.78 | 0.75 | 0.69 | 0.60 | 0.53 | 0.47 | 0.4 | 0.2 |
| 2.20 | 1.80 | 1.38 | 1.27 | 1.21 | 1.15 | 1.10 | 1.06 | 0.96 | 0.94 | 0.91 | 0.88 | 0.85 | 0.79 | 0.70 | 0.63 | 0.57 | 0.5 | 0.3 |
| 2.40 | 1.90 | 1.48 | 1.37 | 1.31 | 1.25 | 1.20 | 1.16 | 1.06 | 1.04 | 1.01 | 0.98 | 0.95 | 0.89 | 0.80 | 0.73 | 0.67 | 0.6 | 0.4 |
| 2.60 | 2.00 | 1.58 | 1.47 | 1.41 | 1.36 | 1.30 | 1.26 | 1.16 | 1.14 | LI I | 1.08 | 1.05 | 0.99 | 0.90 | 0.83 | 0.77 | 0.7 | 0.5 |

*********************************Question Paper Ends Here**********************************